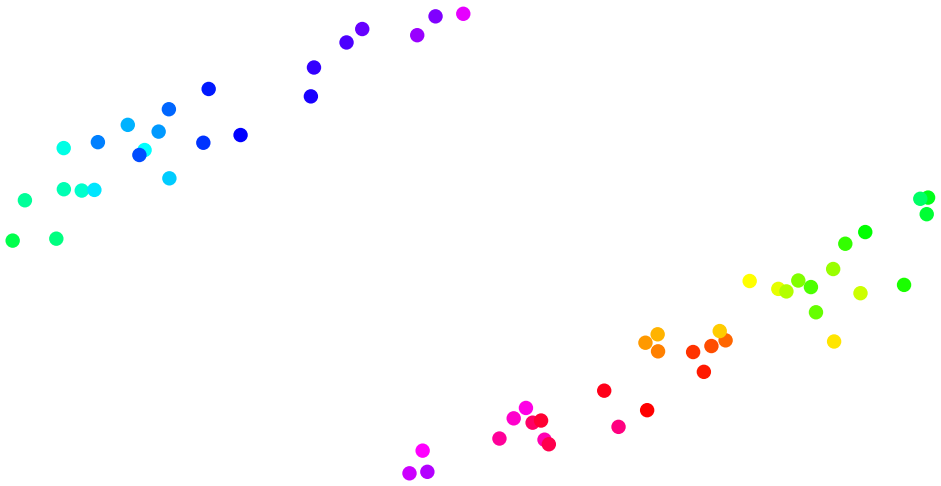
Polysphere Tests

Michael Nisenzon

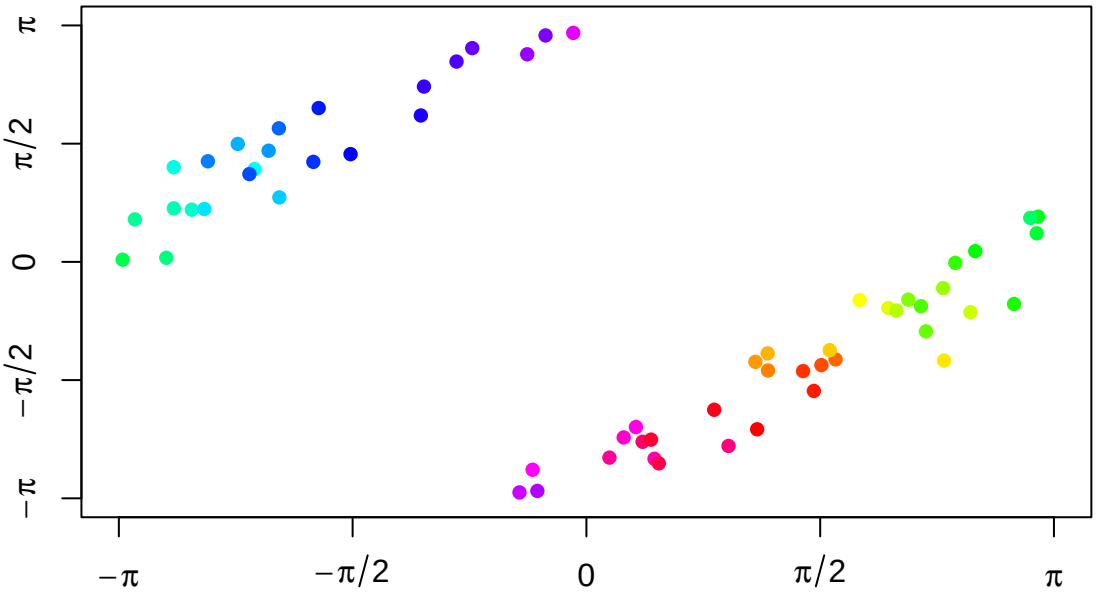
2023-03-14

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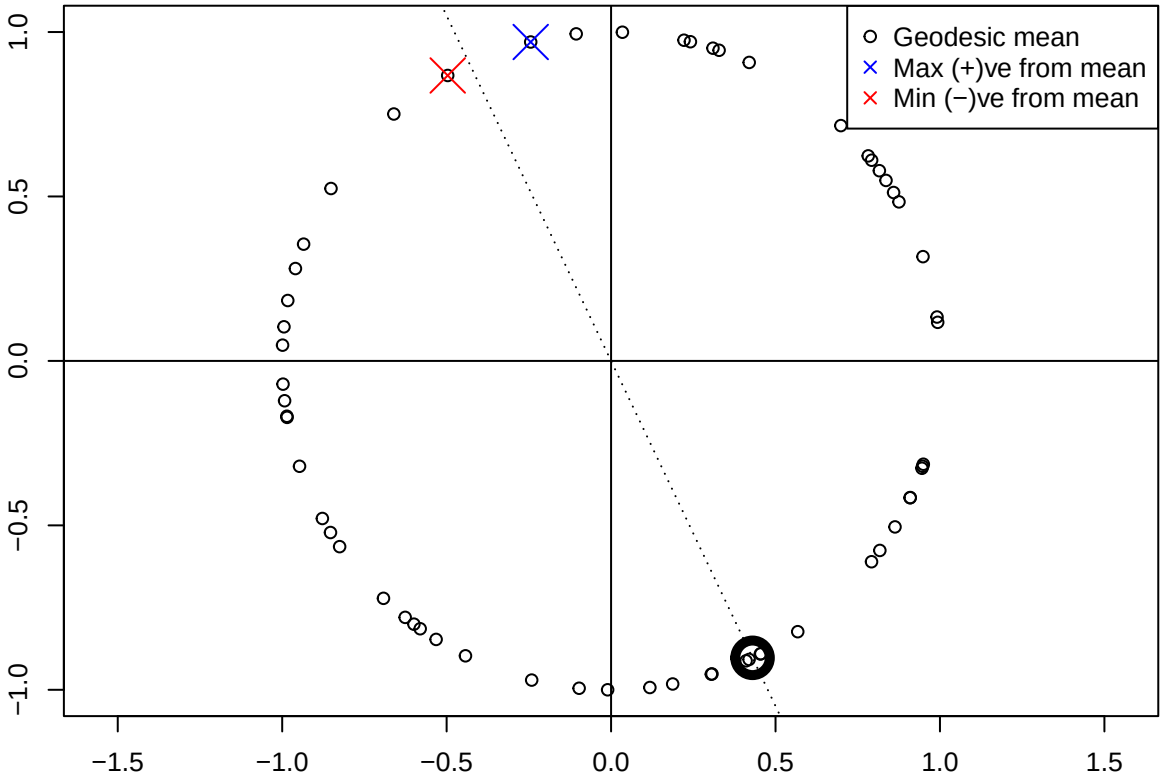
```
#' # Straight trends on (S^1)^2
set.seed(5)
n <- 60
samp_1 <- r_path_s1r(n = n, r = 2, k = c(1, 1), angles = TRUE)
plot(samp_1,
  xlim = c(-pi, pi), ylim = c(-pi, pi), col = rainbow(n),
  axes = FALSE, xlab = "", ylab = "", pch = 16
)
```

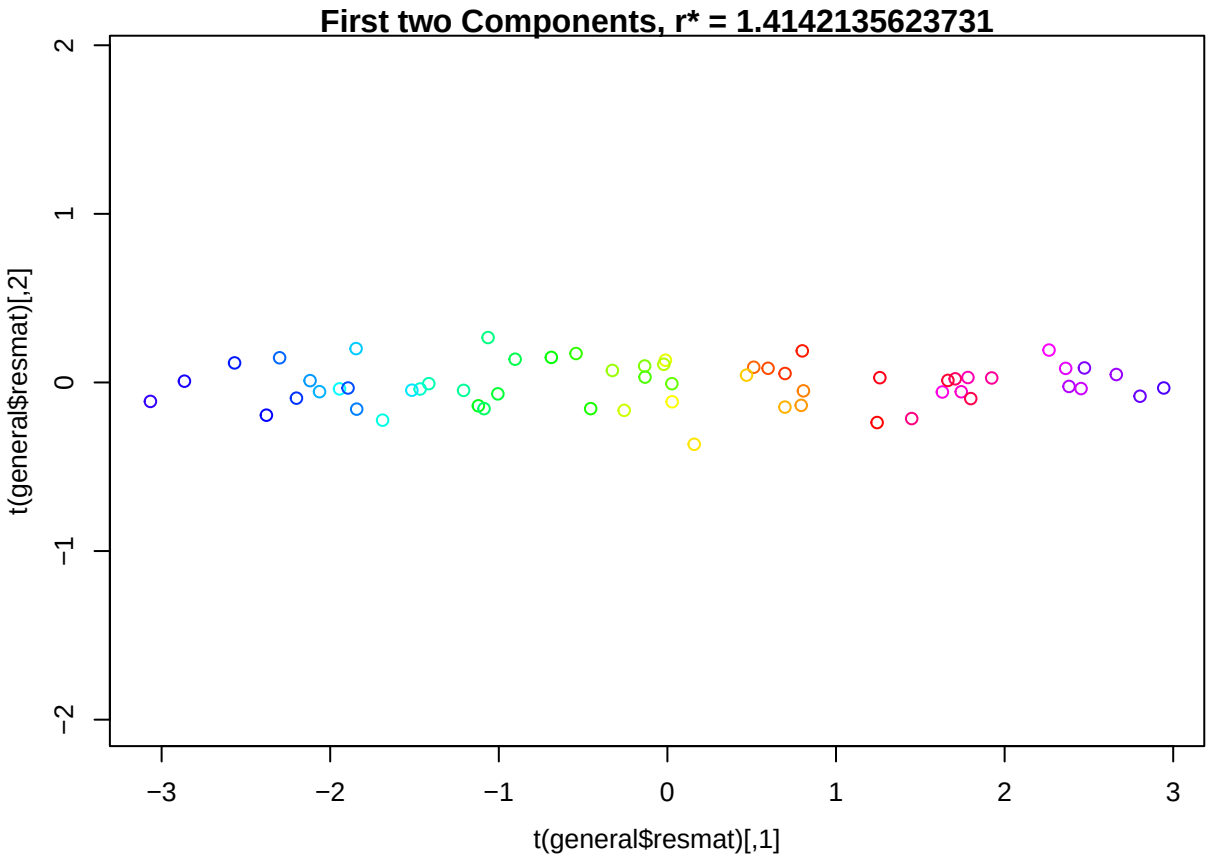


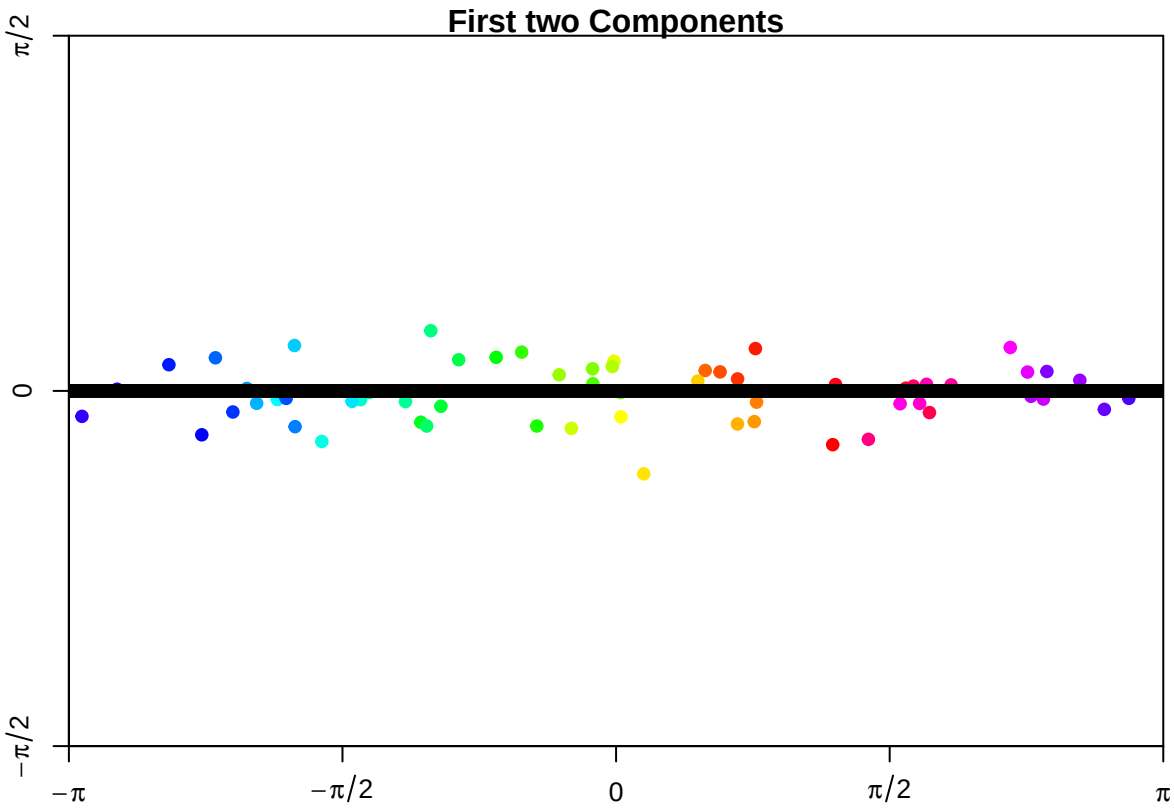
```
sdetorus::torusAxis()
```



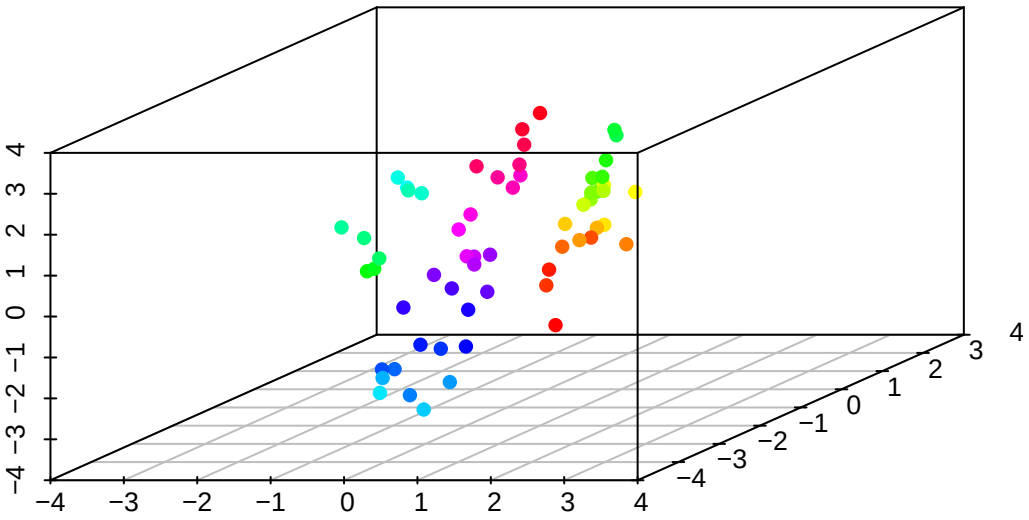

```
clusters <- samp_1
plot_MDS(clusters, angular = TRUE)
```


Geodesic mean
Max (+)ve from mean
Min (−)ve from mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5

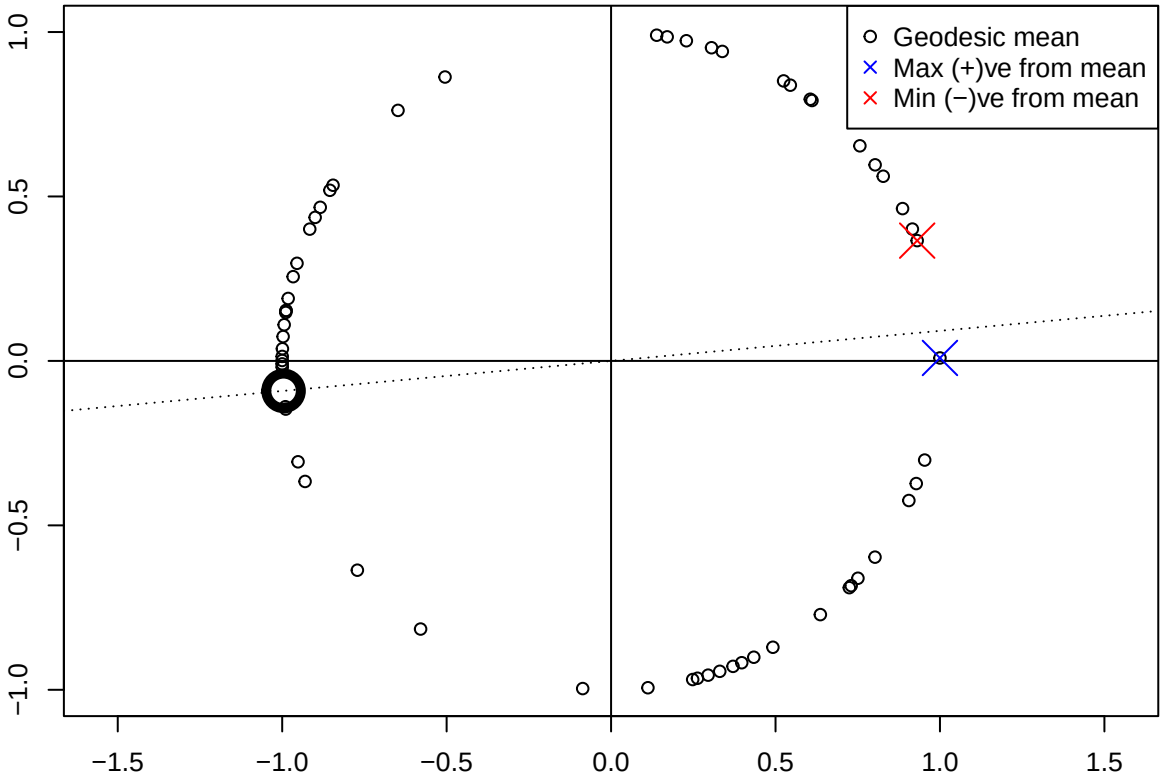

First two Components, r* = 1.4142135623731
t(general$resmat)[,2]
t(general$resmat)[,1]
2
1
0
−1
−2
−3
−2
−1
0
1
2
3

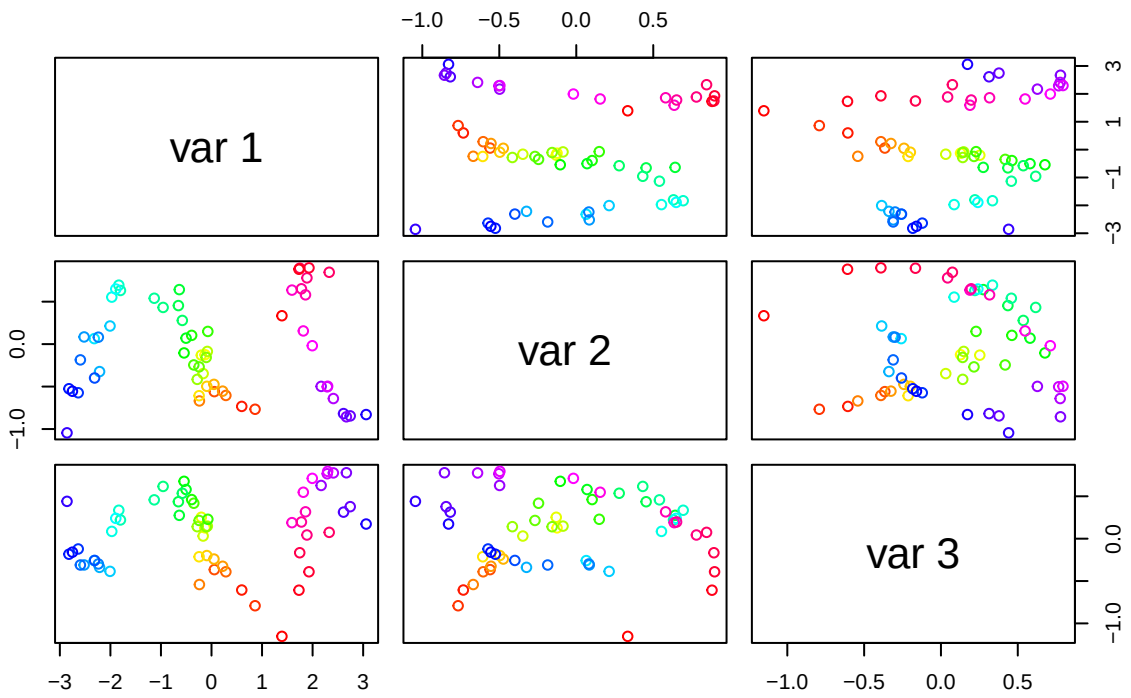


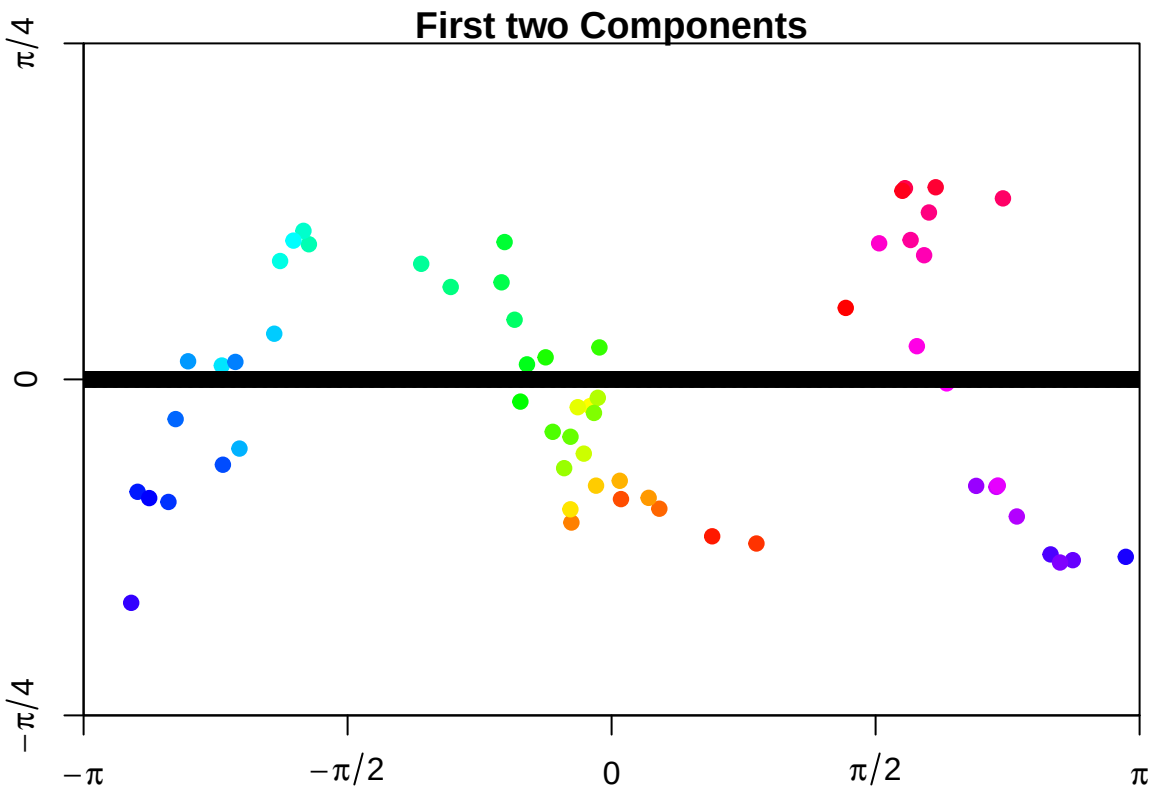

```
#' # Straight trends on (S^1)^3
set.seed(100)
n <- 60
samp_2 <- r_path_s1r(n = n, r = 3, k = c(0, 1, 2), angles = TRUE)
scatterplot3d::scatterplot3d(
  samp_2,
  xlim = c(-pi, pi), ylim = c(-pi, pi), zlim = c(-pi, pi),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
```




```
clusters <- samp_2
invisible(plot_MDS(clusters, angular = TRUE))
```

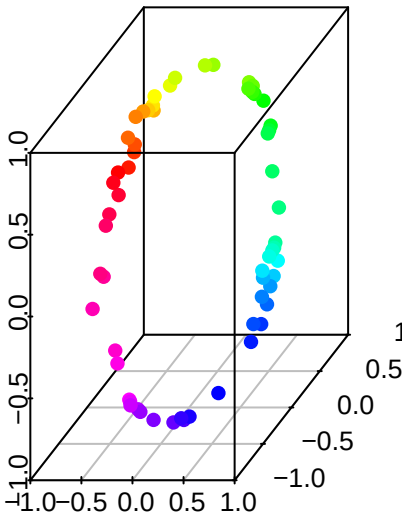

Geodesic mean
Max (+)ve from mean
Min (−)ve from mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5

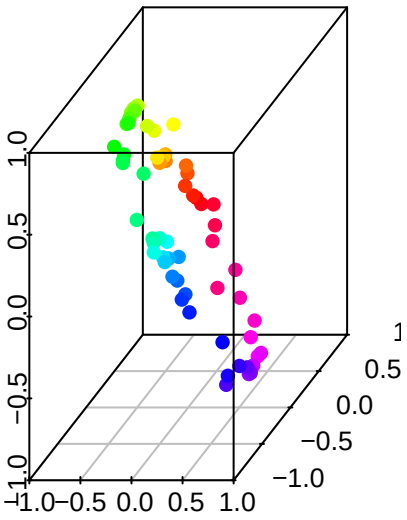

First three Components, r* = 1.73205080756888
var 1
var 2
var 3
−1.0
−0.5
0.0
0.5
3
1
−1
−3
0.0
−1.0
−3
−2
−1
0
1
2
3



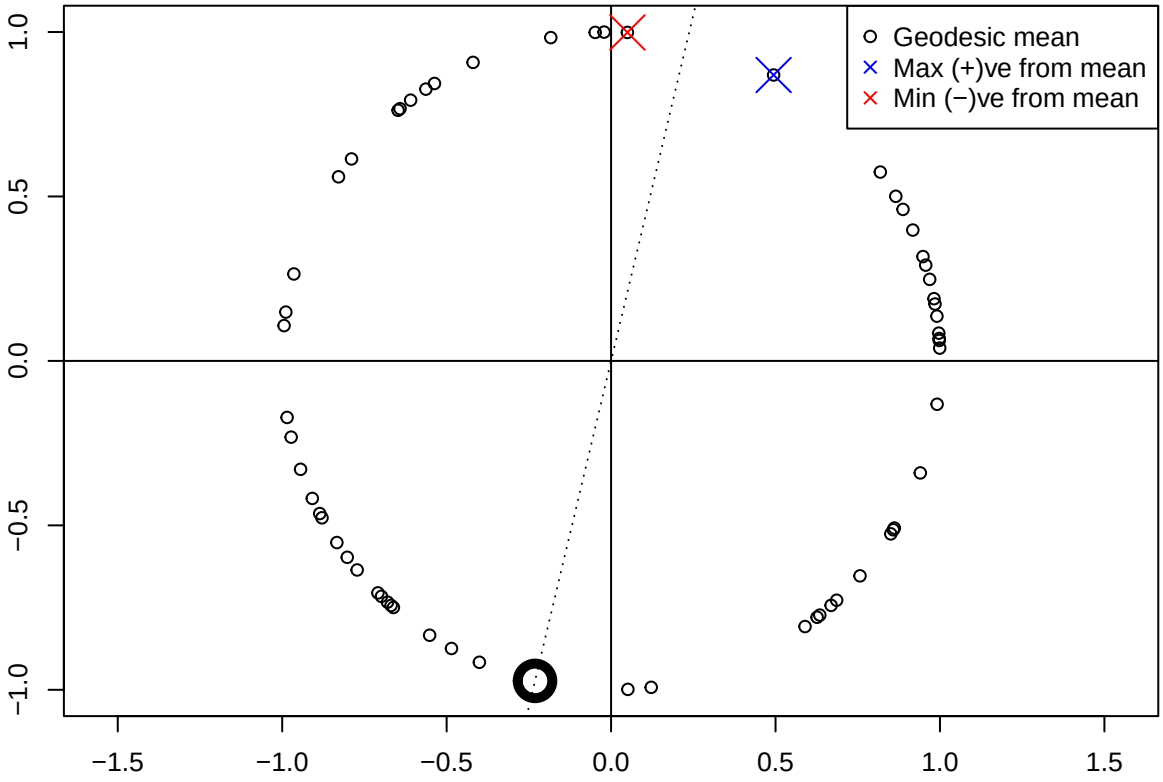

```
#' # Small-circle trends on (S^2)^2
set.seed(20)
n <- 60
samp_3 <- r_path_s2r(n = n, r = 2, sigma = 0.05)
layout(matrix(c(1, 2), 1))

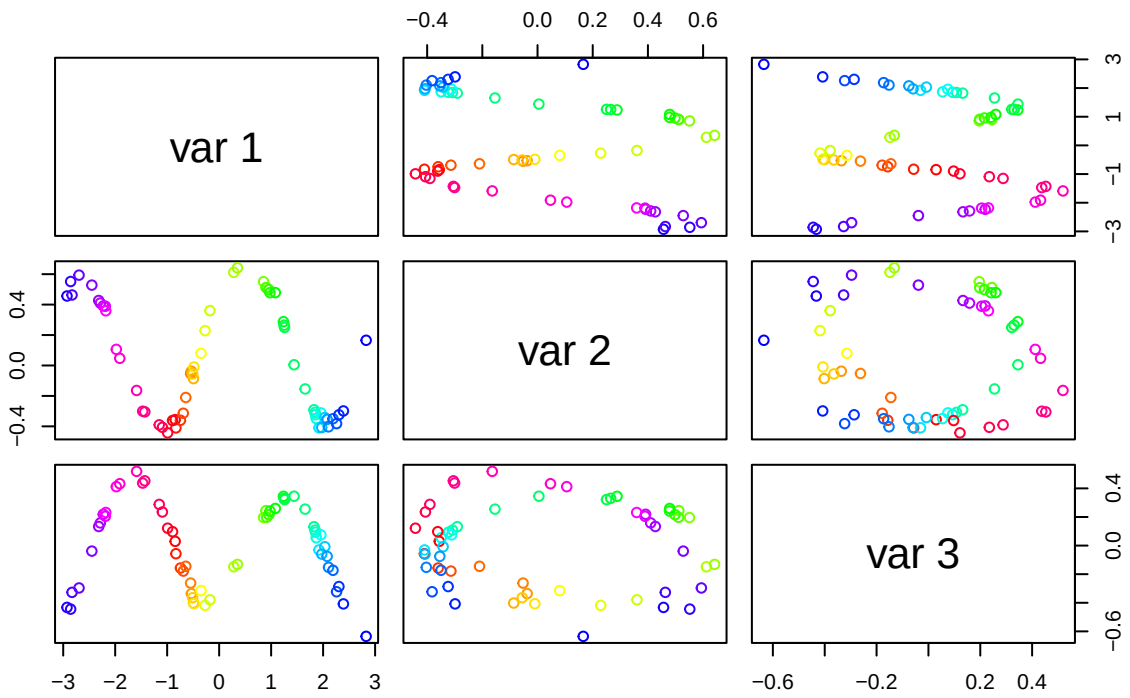
scatterplot3d::scatterplot3d(
  samp_3[, , 1],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
scatterplot3d::scatterplot3d(
  samp_3[, , 2],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
```

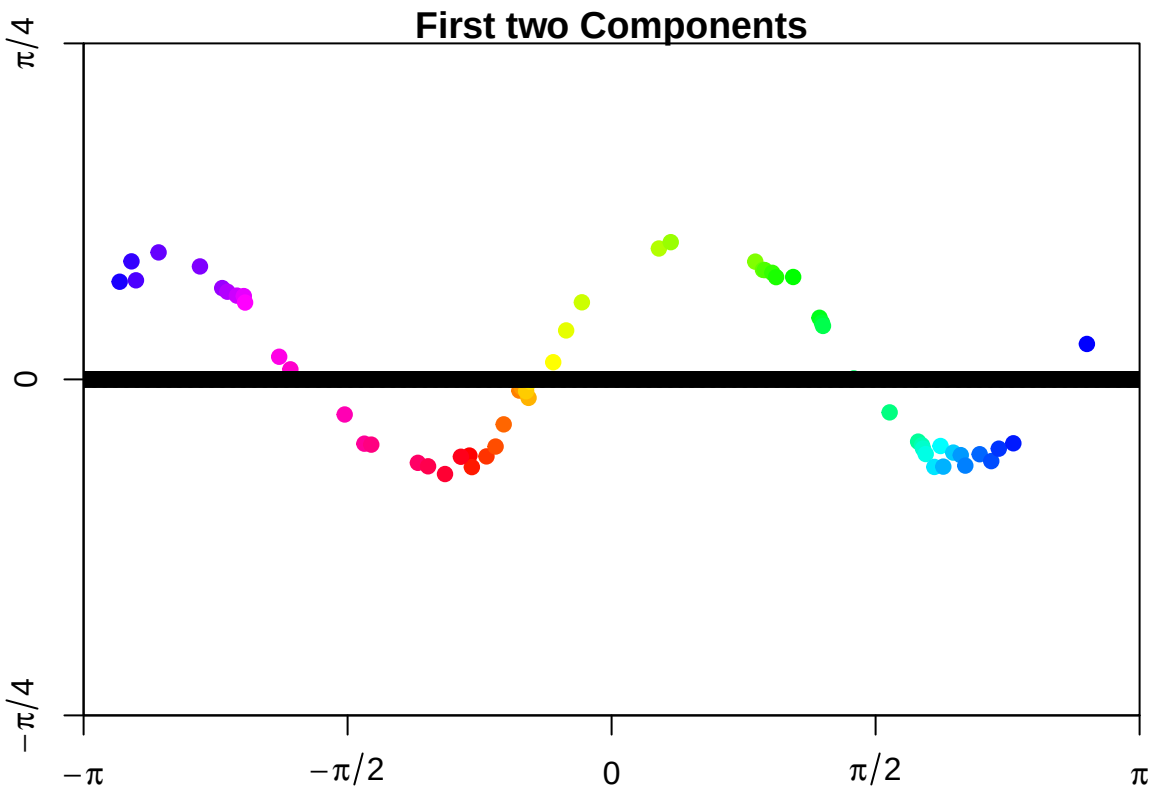





```
clusters <- cbind(samp_3[, , 1], samp_3[, , 2])
plot_MDS(clusters, c(2, 2), angular = TRUE)
```

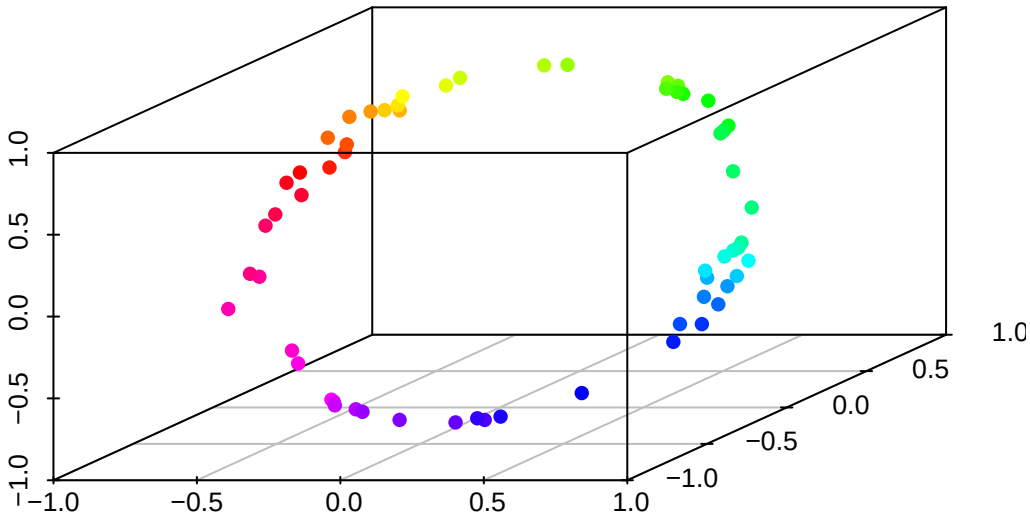

Geodesic mean
Max (+)ve from mean
Min (−)ve from mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5


First three Components, r* = 1.4142135623731
var 1
var 2
var 3
−0.4
0.0
0.2
0.4
0.6
3
1
−1
−3
0.4
0.0
−0.4
−3
−2
−1
0
1
2
3
−0.6
−0.2
0.2
0.4
0.4
0.0
−0.6

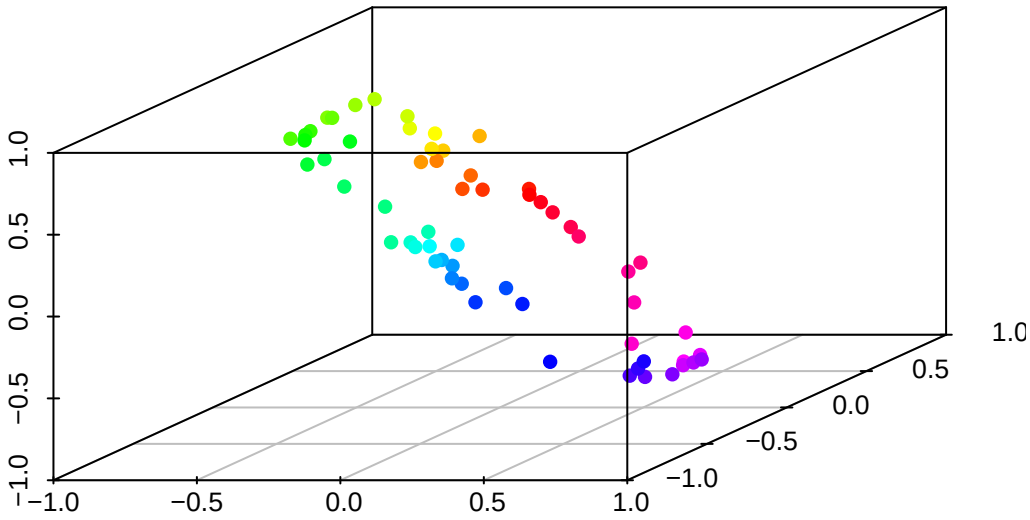



```
#' # Small-circle trends on (S^2)^3
set.seed(20)
n <- 60
samp_3 <- r_path_s2r(n = n, r = 3, sigma = 0.05)

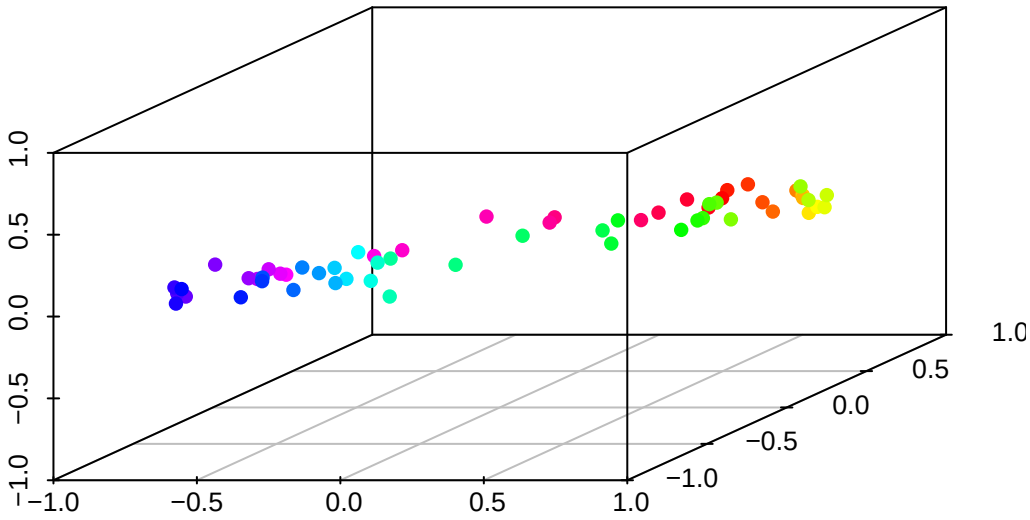
scatterplot3d::scatterplot3d(
  samp_3[, , 1],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
```



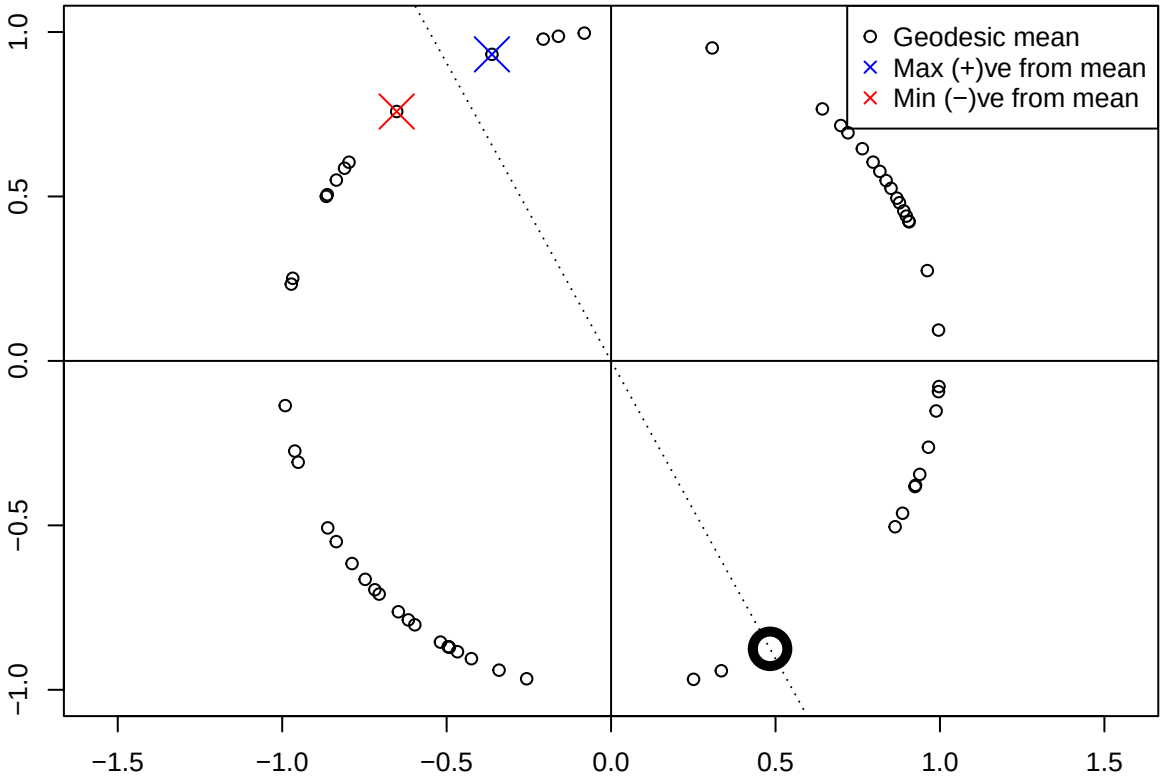

```
scatterplot3d::scatterplot3d(
  samp_3[, , 2],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
```

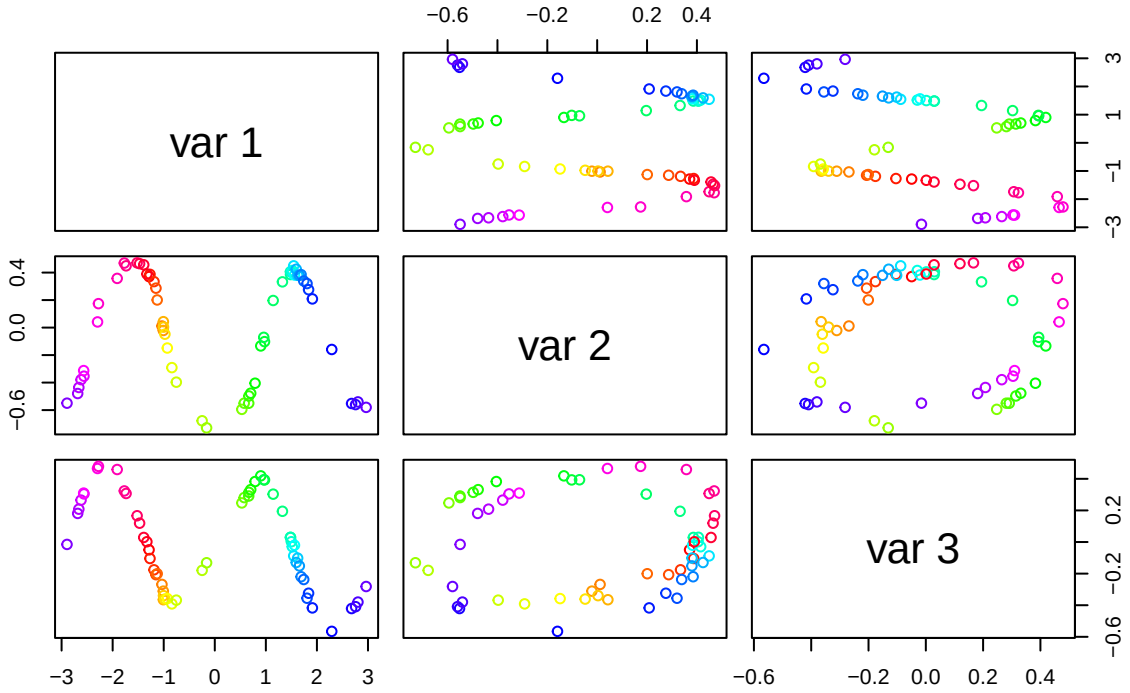


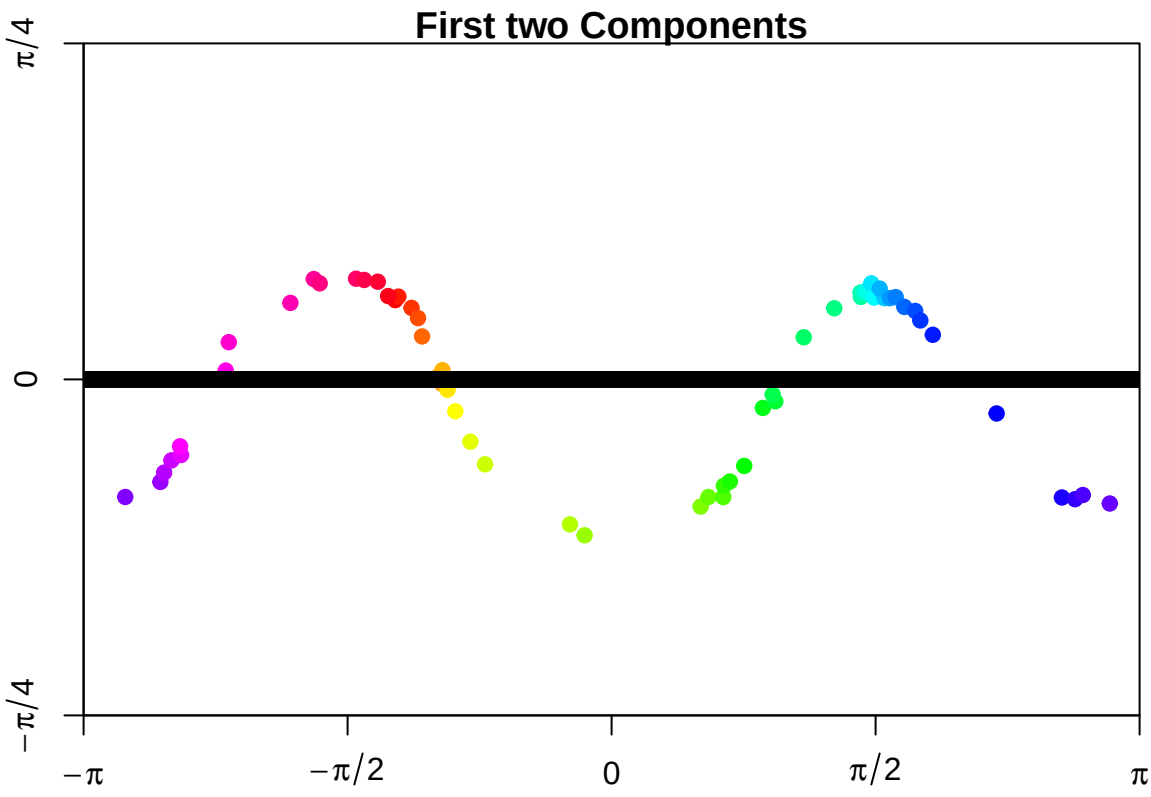

```
scatterplot3d::scatterplot3d(
  samp_3[, , 3],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
```




```
clusters <- cbind(samp_3[, , 1], samp_3[, , 2], samp_3[, , 3])
plot_MDS(clusters, c(2, 2, 2), angular = TRUE)
```


Geodesic mean
Max (+)ve from mean
Min (−)ve from mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5

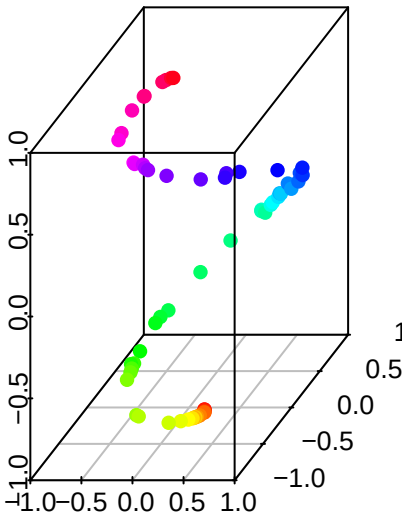

First three Components, r* = 1.73205080756888
var 1
var 2
var 3
−0.6
−0.2
0.2
0.4
3
1
−1
−3
0.4
0.0
−0.6
−3
−2
−1
0
1
2
3
−0.6
−0.2
0.0
0.2
0.4
0.2
−0.2
−0.6

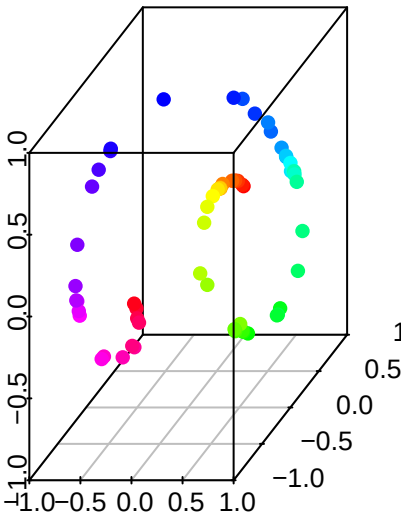



```
#' # Spiral trends on (S^2)^2
set.seed(20)
n <- 60

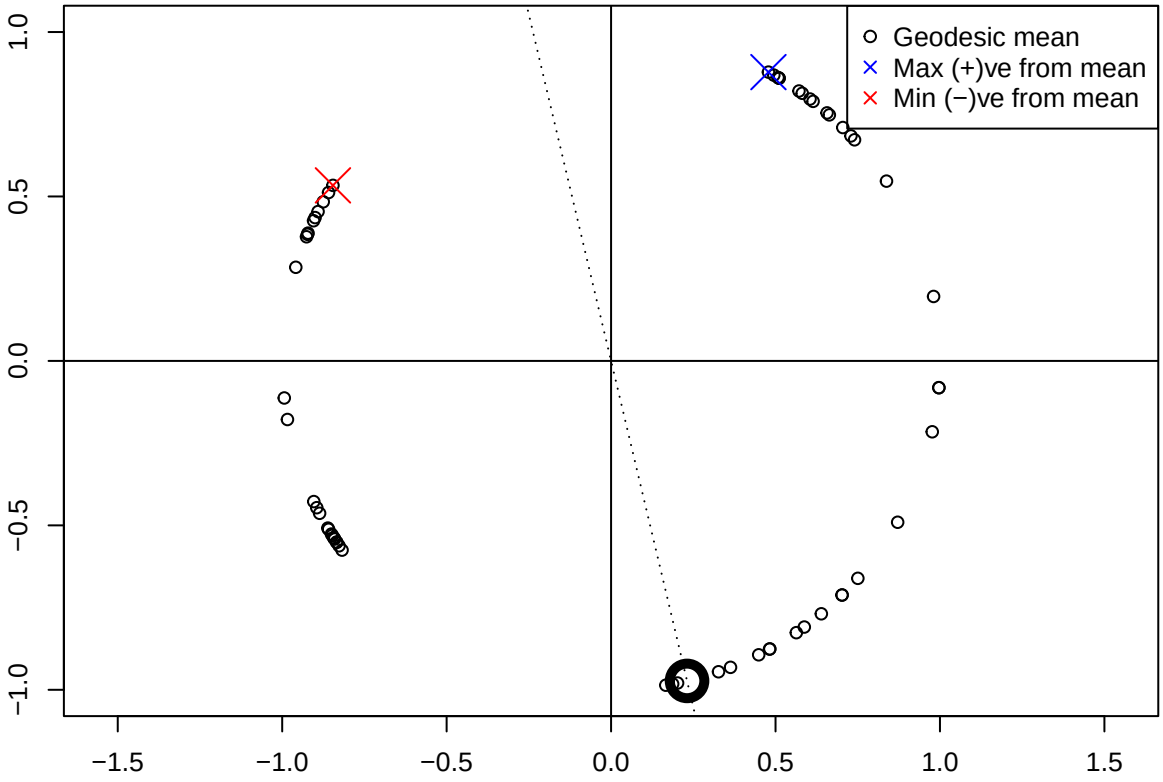
samp_4 <- r_path_s2r(n = n, r = 2, c = 3, spiral = TRUE, sigma = 0.05)
layout(matrix(c(1, 2), 1))

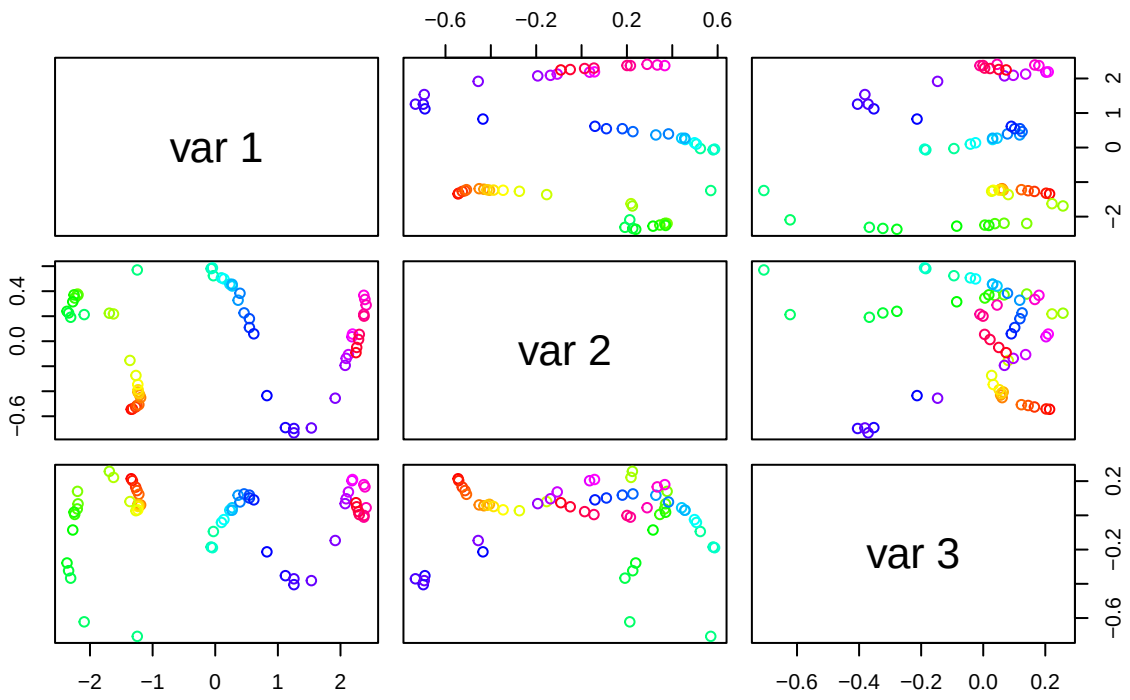
scatterplot3d::scatterplot3d(
  samp_4[, , 1],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
scatterplot3d::scatterplot3d(
  samp_4[, , 2],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
```

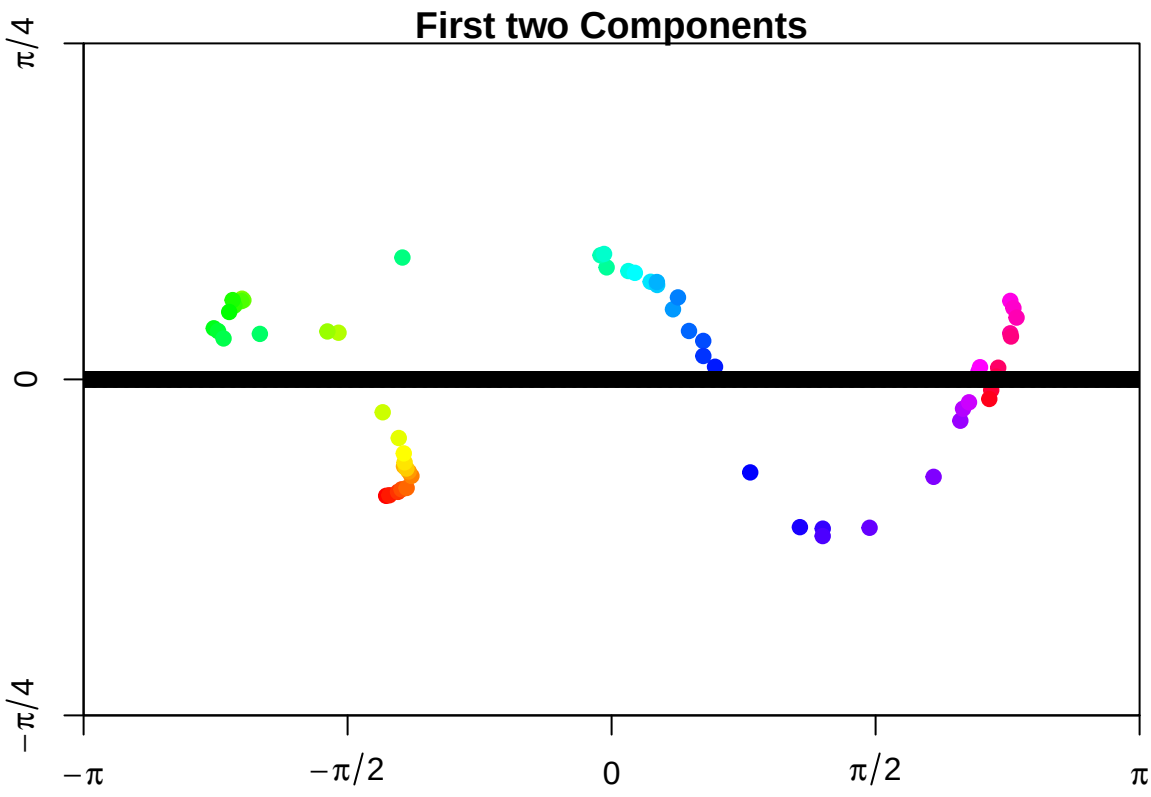





```
clusters <- cbind(samp_4[, , 1], samp_4[, , 2])
plot_MDS(clusters, c(2, 2), angular = TRUE)
```


Geodesic mean
Max (+)ve from mean
Min (−)ve from mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5

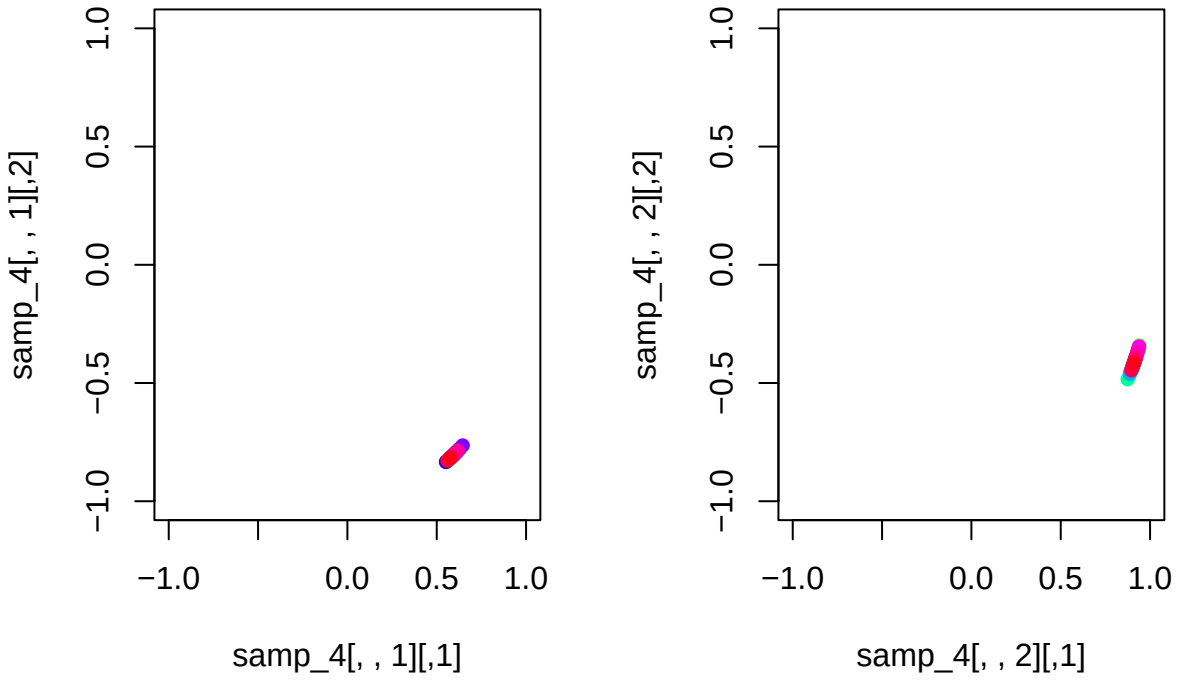

First three Components, r* = 1.4142135623731
var 1
var 2
var 3
−0.6
−0.2
0.2
0.6
−2
0
1
2
0.4
0.0
−0.6
−2
−1
0
1
2
−0.6
−0.4
−0.2
0.0
0.2
0.2
−0.2
−0.6



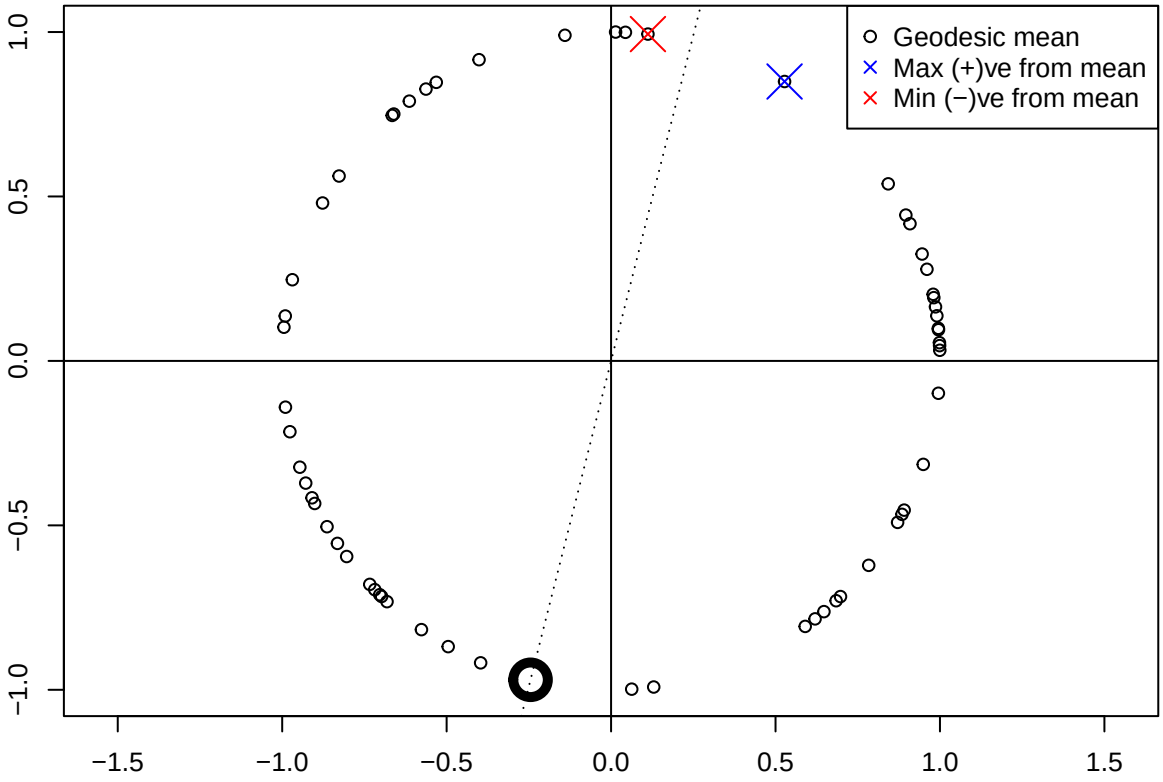
3/21/2023

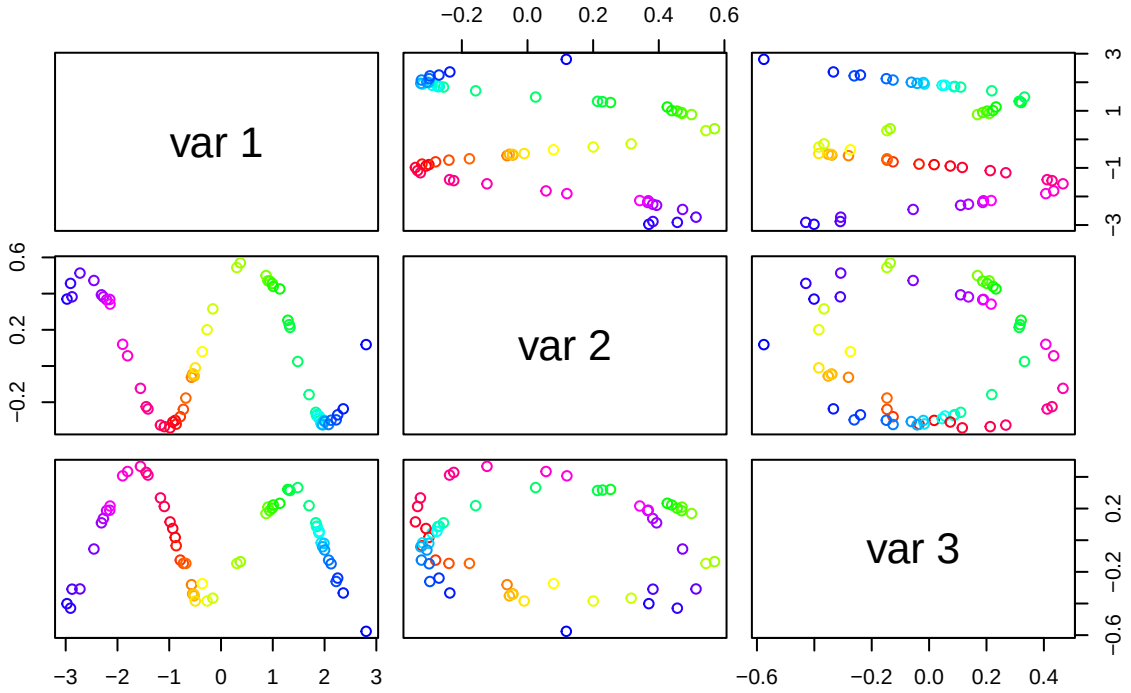
```
#' # Small circle trends on (S^2)^2 times (S^1)^3, almost all variance along first two spheres.
set.seed(20)
n <- 60
samp_3 <- r_path_s2r(n = n, r = 2, sigma = 0.05)
samp_4 <- r_path_s1r(n = n, r = 3, k = rep(0, 3), sigma = 0.03)
layout(matrix(c(1, 2), 1))

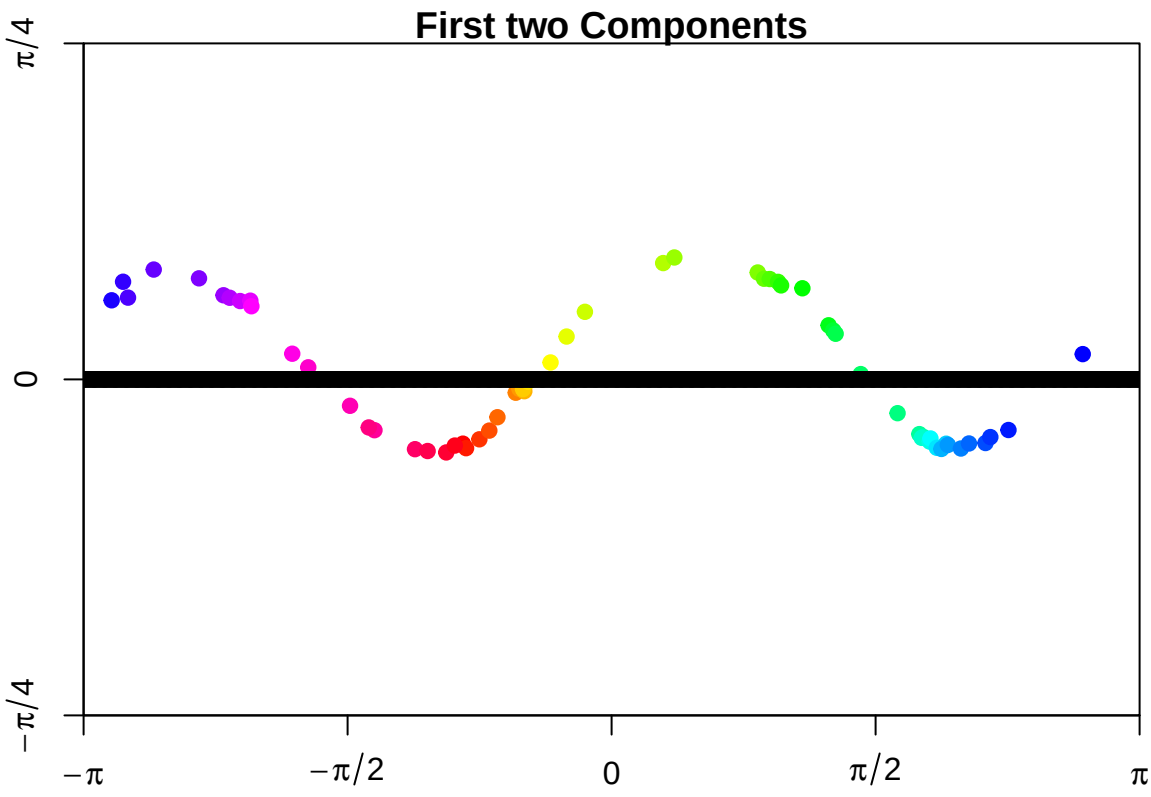
plot( samp_4[, , 1],xlim = c(-1,1), ylim = c(-1,1),col = rainbow(n),pch=16)
plot( samp_4[, , 2],xlim = c(-1,1), ylim = c(-1,1),col = rainbow(n),pch=16)
```




```
clusters <- cbind(samp_3[, , 1], samp_3[, , 2], samp_4[, , 1], samp_4[, , 2], samp_4[, , 3])
plot_MDS(clusters, c(2, 2, 1, 1,1), angular = TRUE)
```


Geodesic mean
Max (+)ve from mean
Min (−)ve from mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5

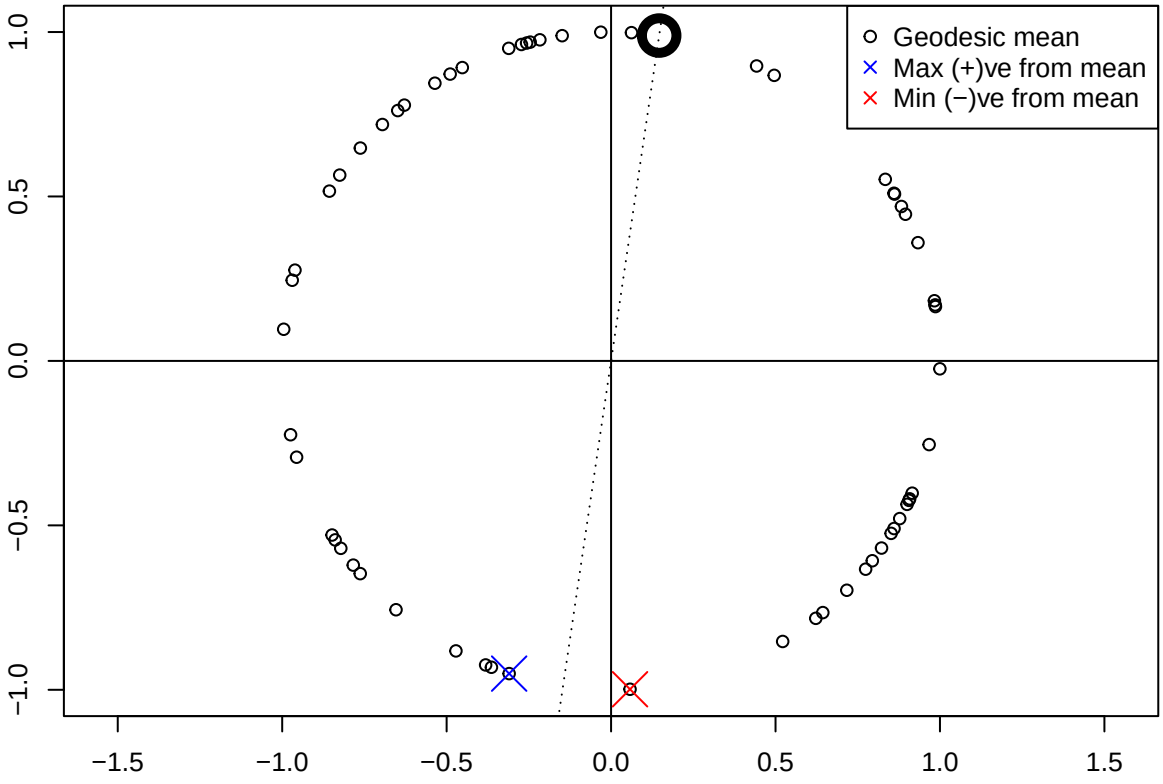

First three Components, r* = 2.23606797749979
var 1
var 2
var 3
−0.2
0.0
0.2
0.4
0.6
3
1
−1
−3
0.6
0.2
−0.2
−3
−2
−1
0
1
2
3
−0.6
−0.2
0.0
0.2
0.4

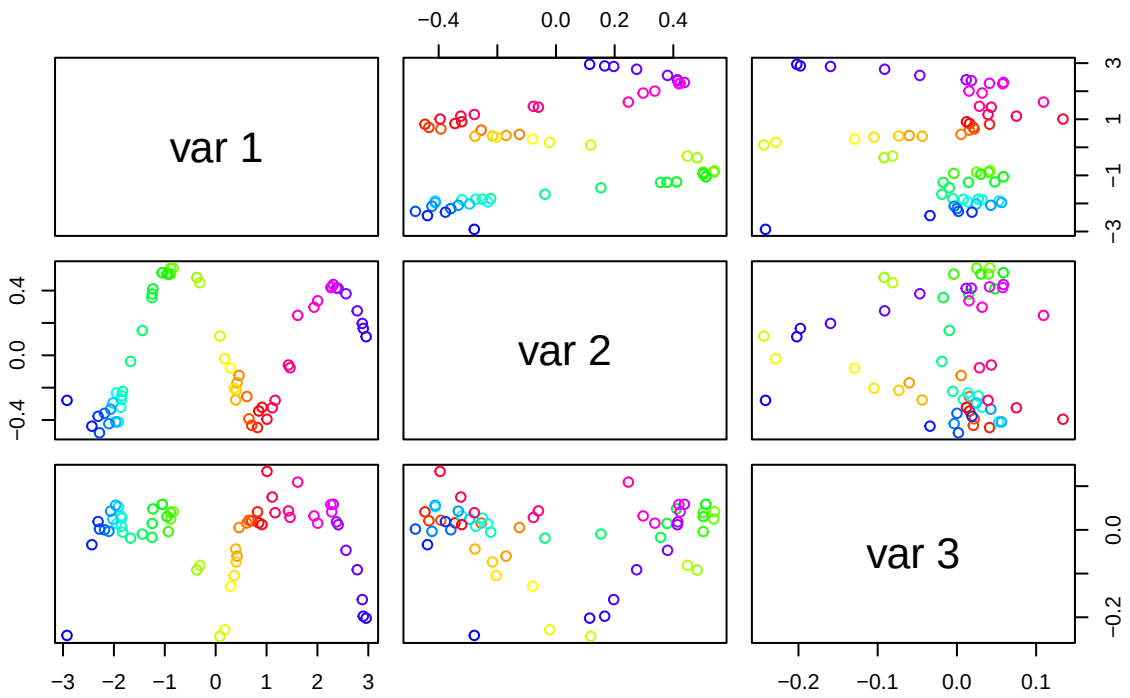


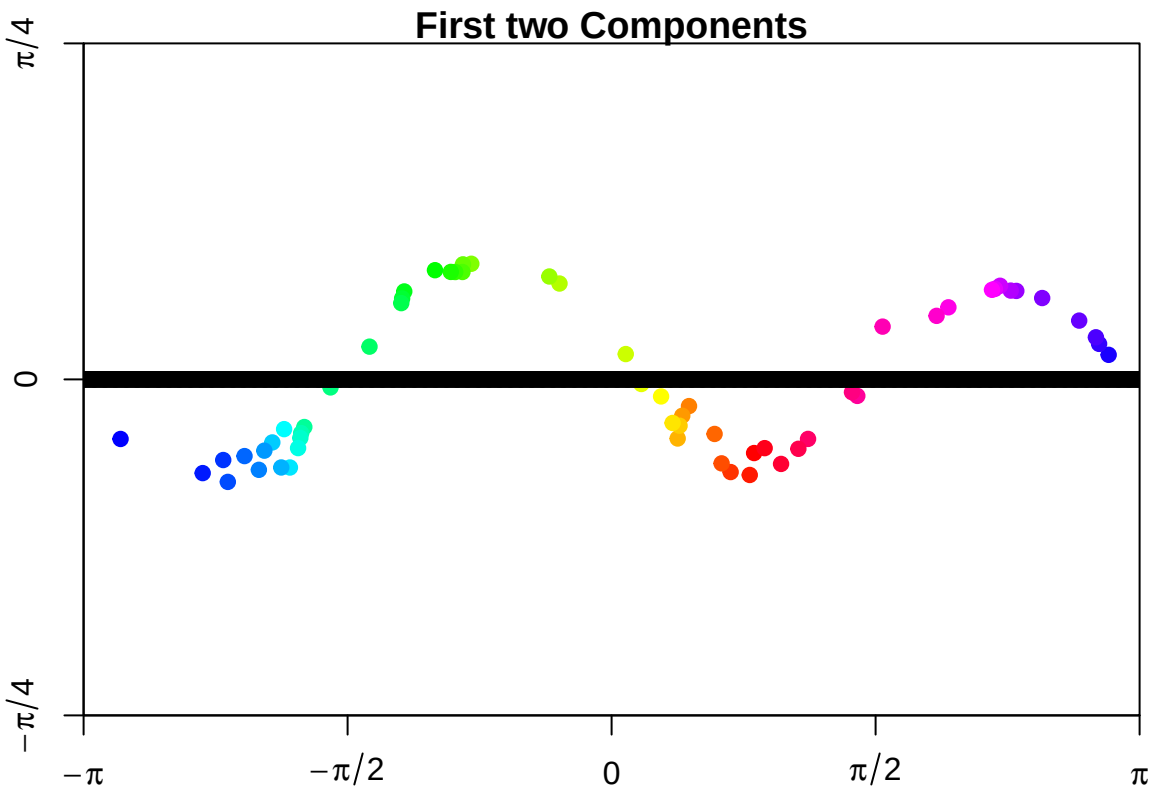

```
#' # Small circle trends on (S^2)^3, almost all variance along first sphere.
set.seed(20)
n <- 60
nspheres <- 3
k<-500
samp_3 <- r_path_s2r(n=n, r = 1, sigma = 0.05)
clusters <- samp_3[, , 1]

for (i in c(2:nspheres)) {
  clusters <- cbind(clusters,rotasym::r_vMF(n,c(rotasym::r_unif_sphere(1,3)),k))

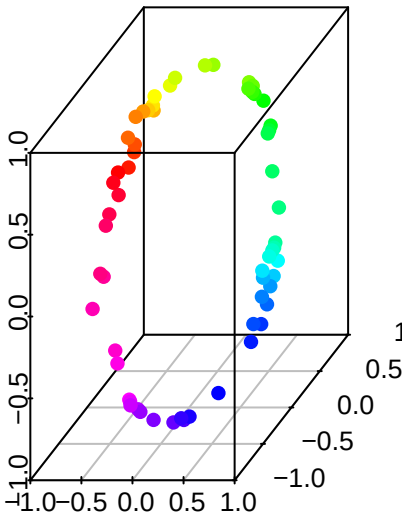
}
plot_MDS(clusters, rep(2,nspheres), angular = TRUE)
```

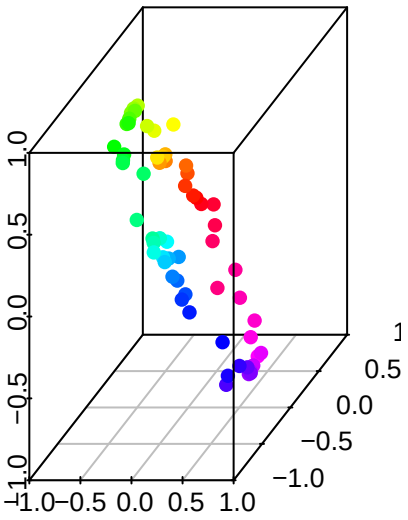

Geodesic mean
Max (+)ve from mean
Min (−)ve from mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5


First three Components, r* = 1.73205080756888
var 1
var 2
var 3
−0.4
0.0
0.2
0.4
3
1
−1
−3
0.4
0.0
−0.4
−3
−2
−1
0
1
2
3
−0.2
−0.1
0.0
0.1
0.0
−0.2




```
#' # Small circle trends on (S^2)^4, almost all variance along first two spheres.
set.seed(20)
n <- 60
nspheres <- 4
k<- 100
samp_3 <- r_path_s2r(n=n, r = 2, sigma = 0.05)
layout(matrix(c(1, 2), 1))
scatterplot3d::scatterplot3d(
  samp_3[, , 1],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
scatterplot3d::scatterplot3d(
  samp_3[, , 2],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
```

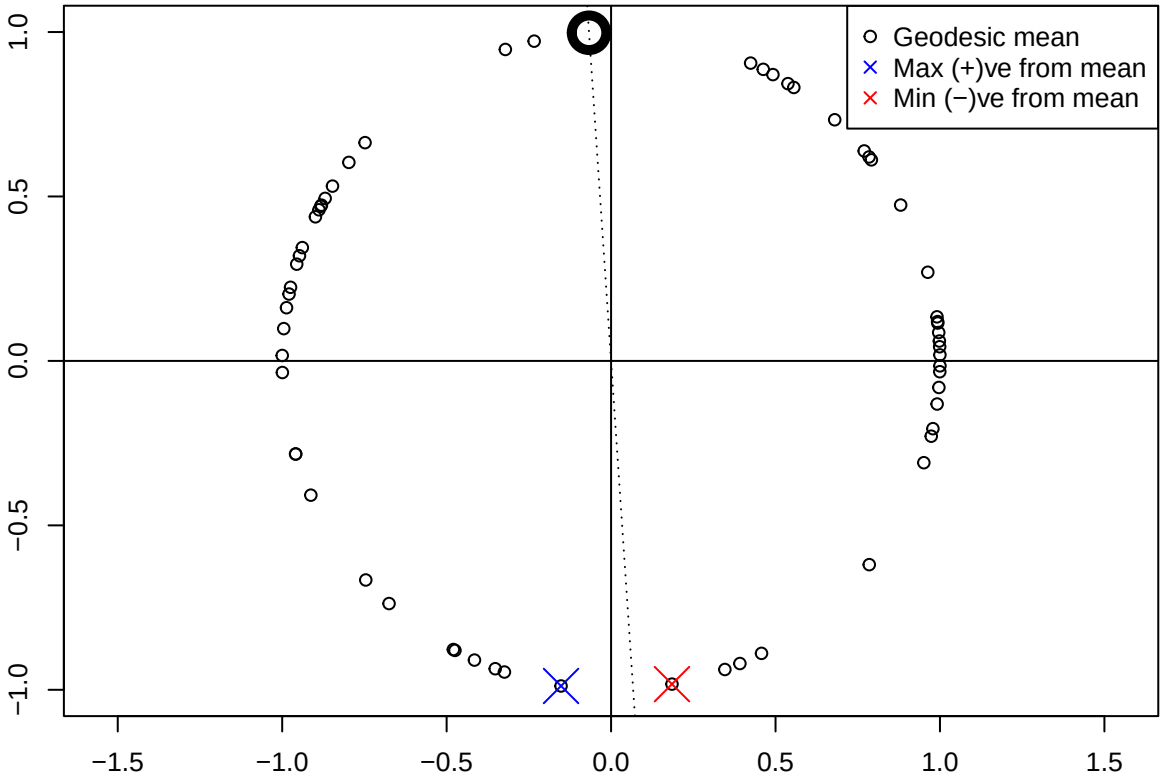


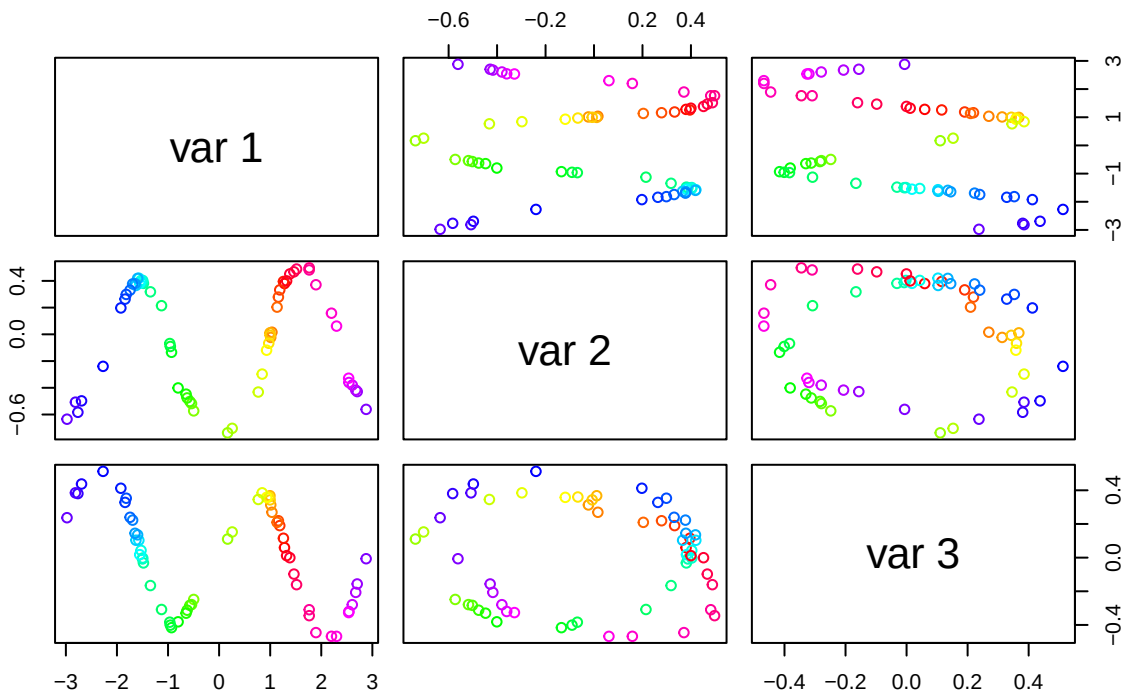


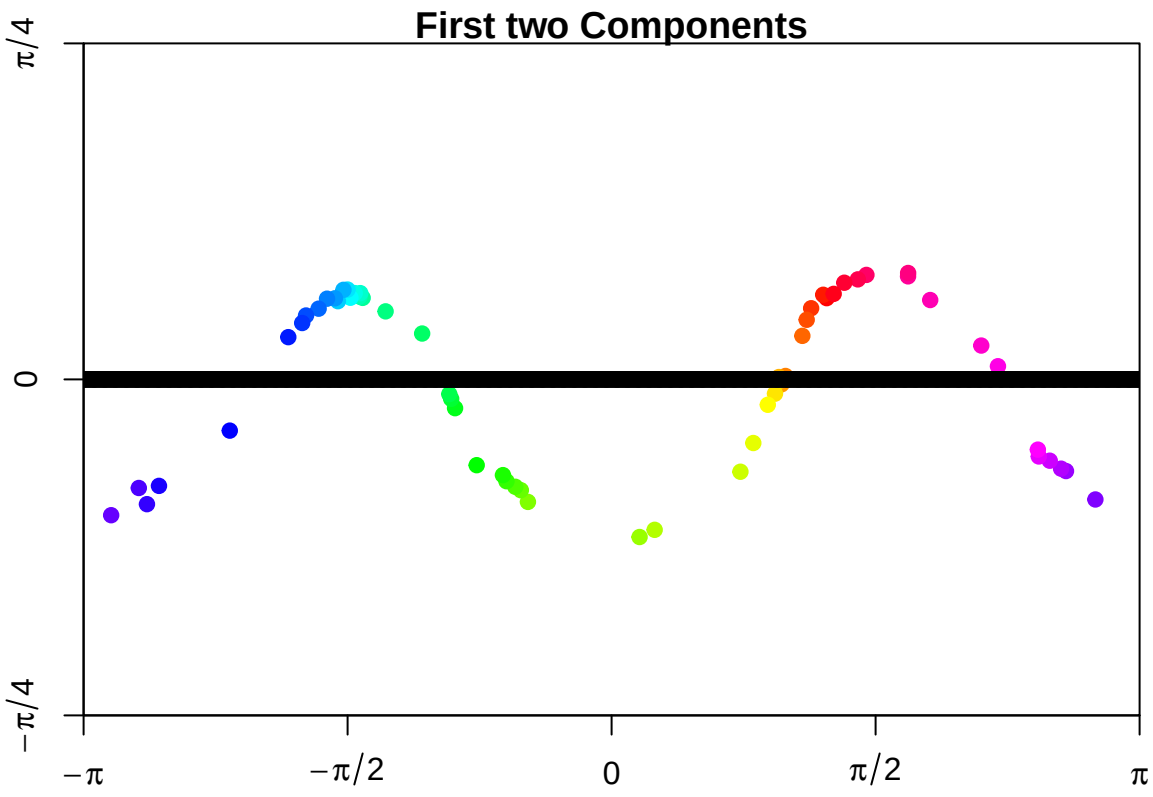

```
clusters <- cbind(samp_3[, , 1],samp_3[, , 2])

for (i in c(2:nspheres)) {
  clusters <- cbind(clusters,rotasym::r_vMF(n,c(rotasym::r_unif_sphere(1,3)),k))

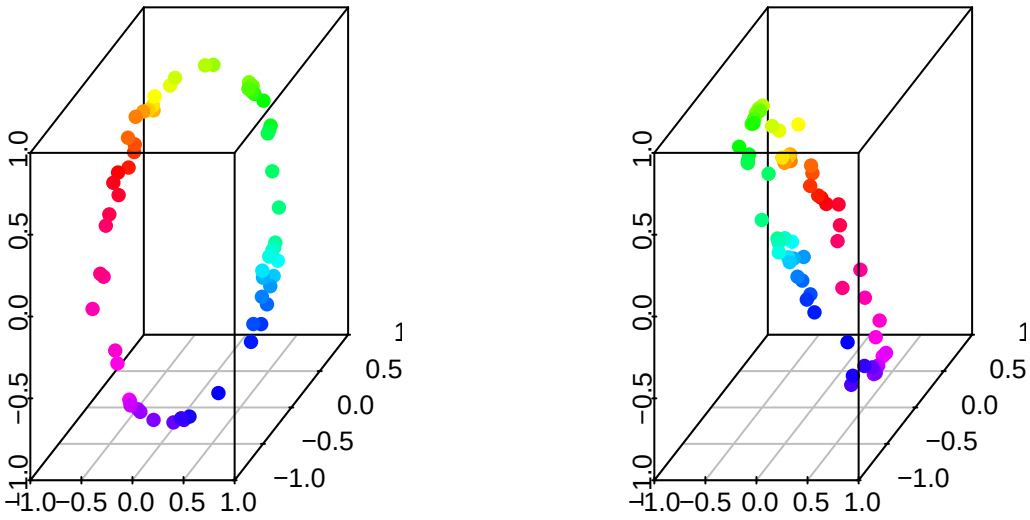
}
plot_MDS(clusters, rep(2,nspheres), angular = TRUE)
```


Geodesic mean
Max (+)ve from mean
Min (−)ve from mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5


First three Components, r* = 2
var 1
var 2
var 3
−0.6
−0.2
0.2
0.4
3
1
−1
−3
0.4
0.0
−0.6
−3
−2
−1
0
1
2
3
−0.4
−0.2
0.0
0.2
0.4



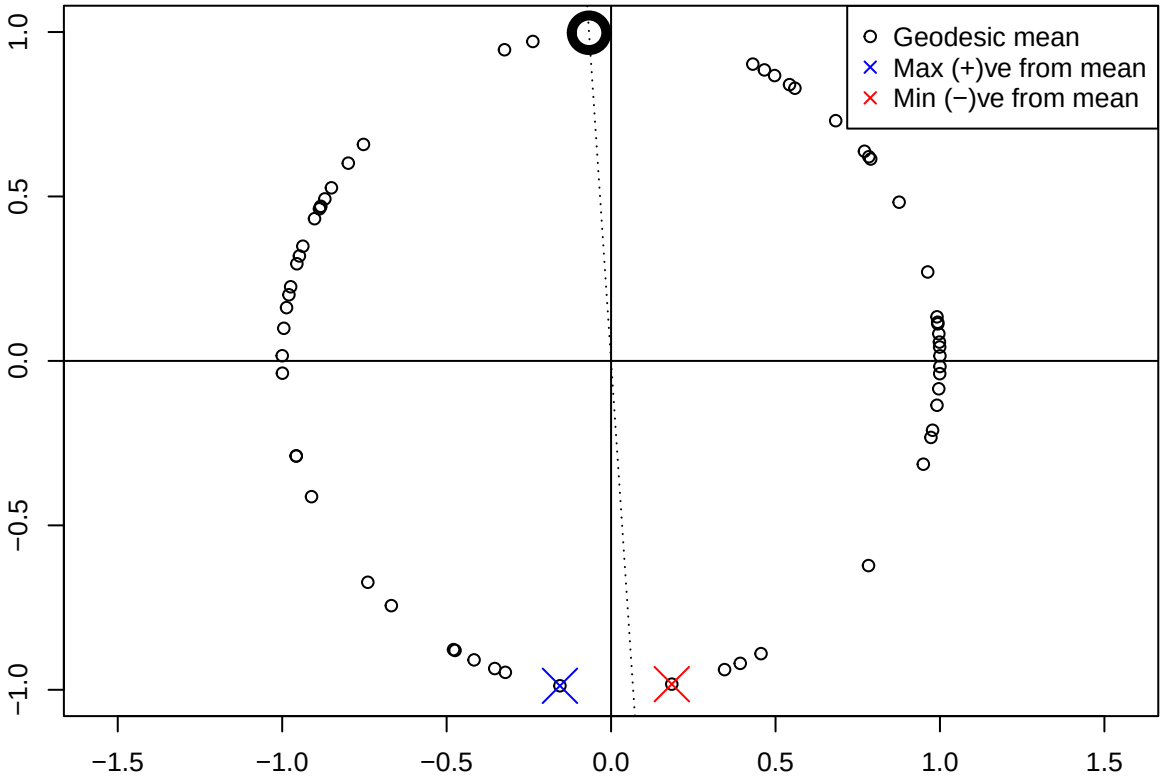

```
#' # Small circle trends on (S^2)^6, almost all variance along first two spheres.
set.seed(20)
n <- 60
nspheres <- 6
k<- 200
samp_3 <- r_path_s2r(n=n, r = 2, sigma = 0.05)
layout(matrix(c(1, 2), 1))
scatterplot3d::scatterplot3d(
  samp_3[, , 1],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
scatterplot3d::scatterplot3d(
  samp_3[, , 2],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
```

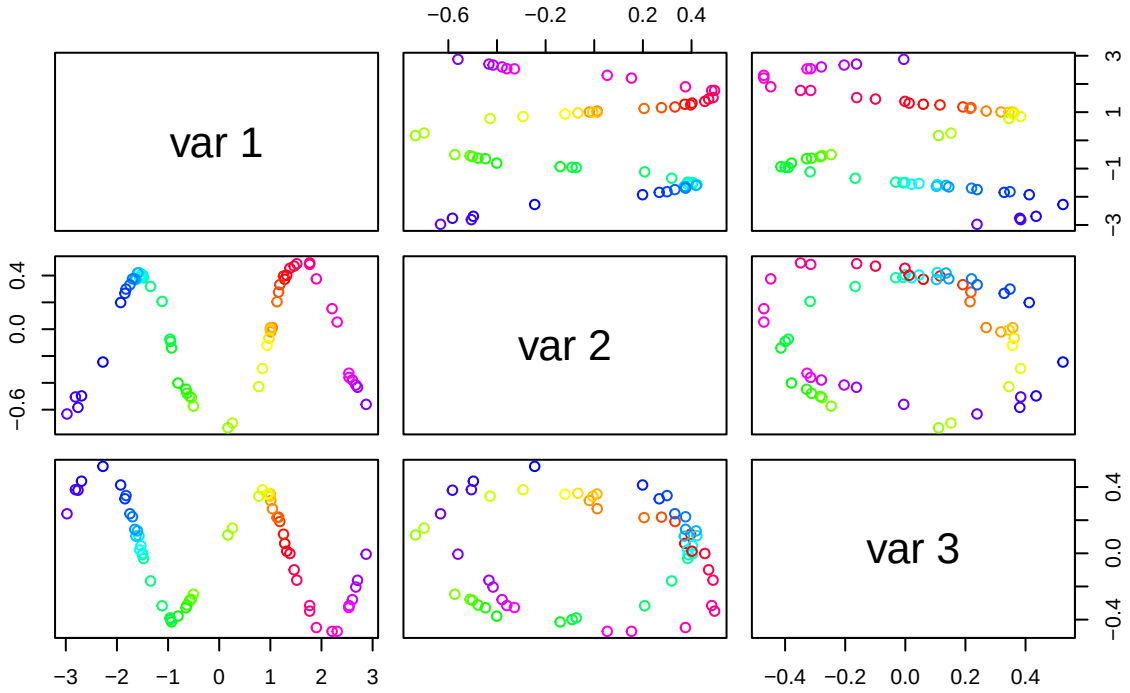


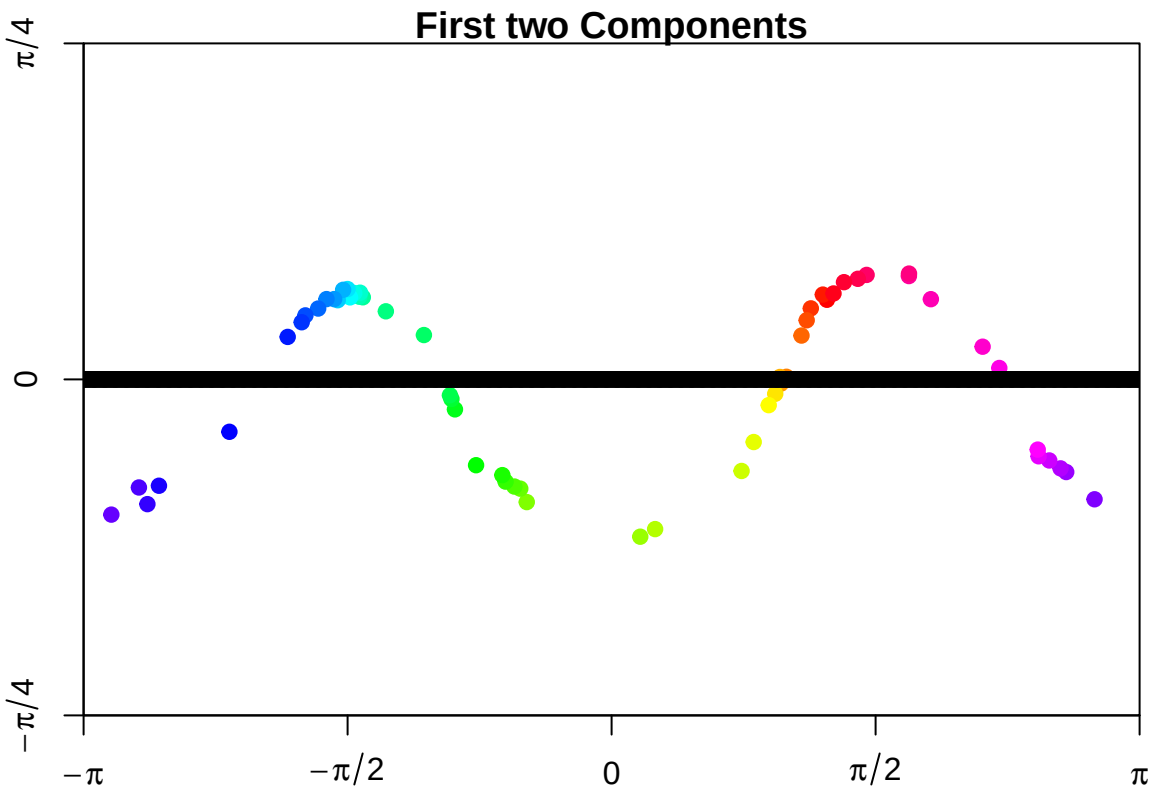

```
clusters <- cbind(samp_3[, , 1],samp_3[, , 2])

for (i in c(2:nspheres)) {
  clusters <- cbind(clusters,rotasym::r_vMF(n,c(rotasym::r_unif_sphere(1,3)),k))

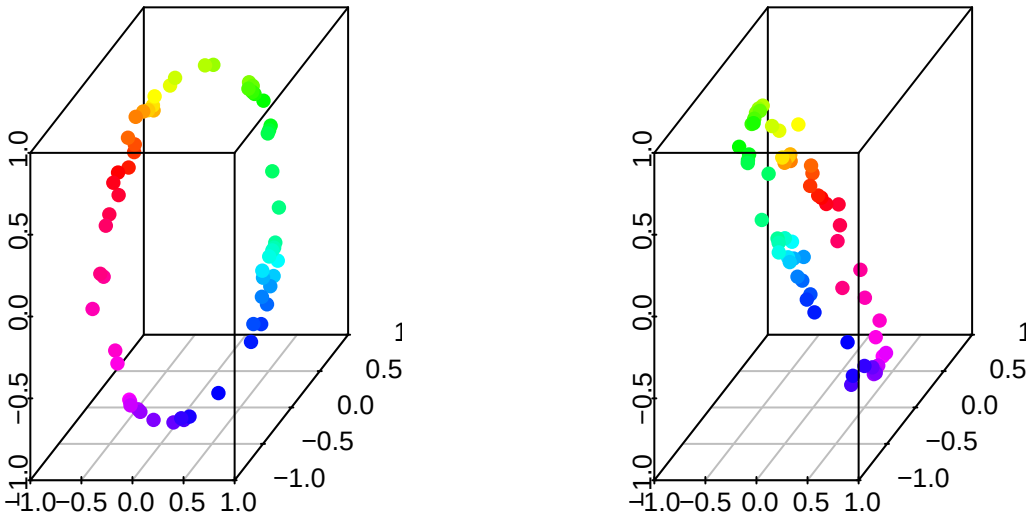
}
plot_MDS(clusters, rep(2,nspheres), angular = TRUE)
```


Geodesic mean
Max (+)ve from mean
Min (−)ve from mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5


First three Components, r* = 2.44948974278318
var 1
var 2
var 3
−0.6
−0.2
0.2
0.4
3
1
−1
−3
0.4
0.0
−0.6
−3
−2
−1
0
1
2
3
−0.4
−0.2
0.0
0.2
0.4



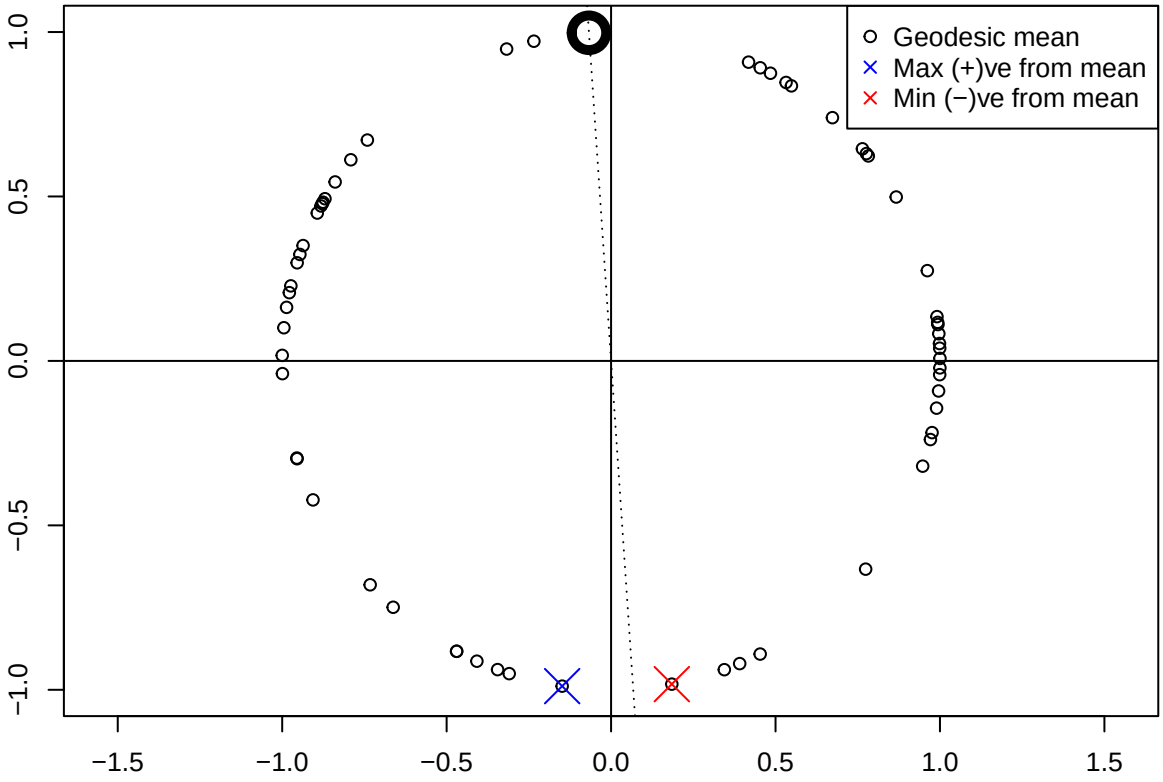

```
#' # Small circle trends on (S^2)^10, almost all variance along first two spheres.
set.seed(20)
n <- 60
nspheres <- 10
k<- 500
samp_3 <- r_path_s2r(n=n, r = 2, sigma = 0.05)
layout(matrix(c(1, 2), 1))
scatterplot3d::scatterplot3d(
  samp_3[, , 1],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
scatterplot3d::scatterplot3d(
  samp_3[, , 2],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
```

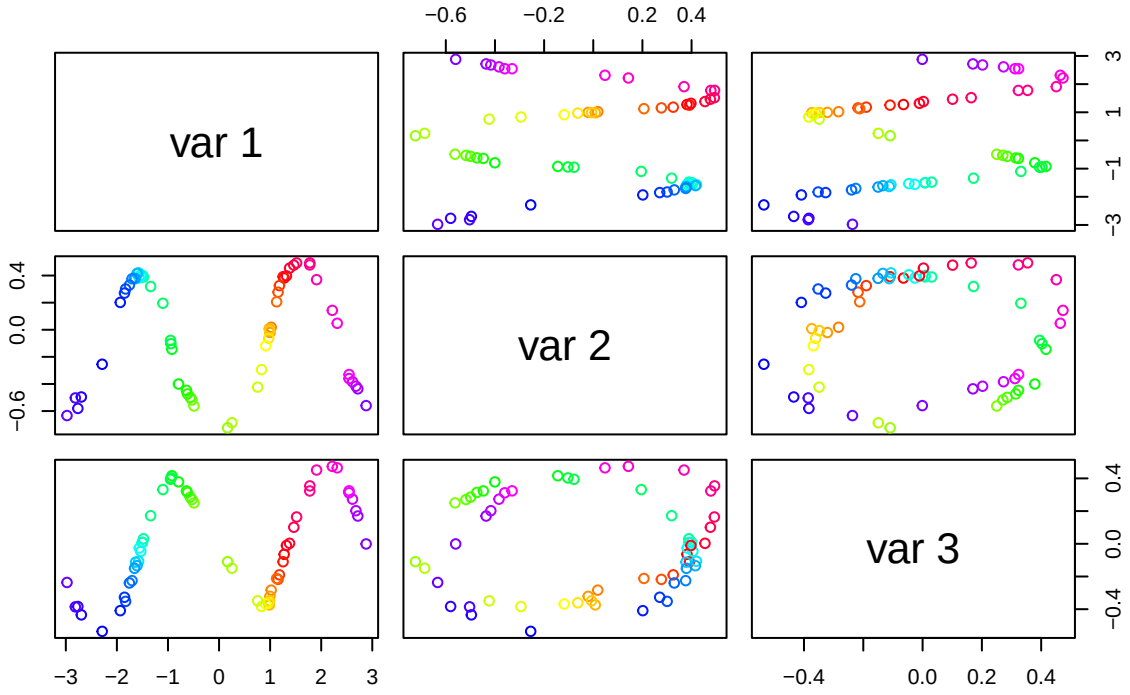


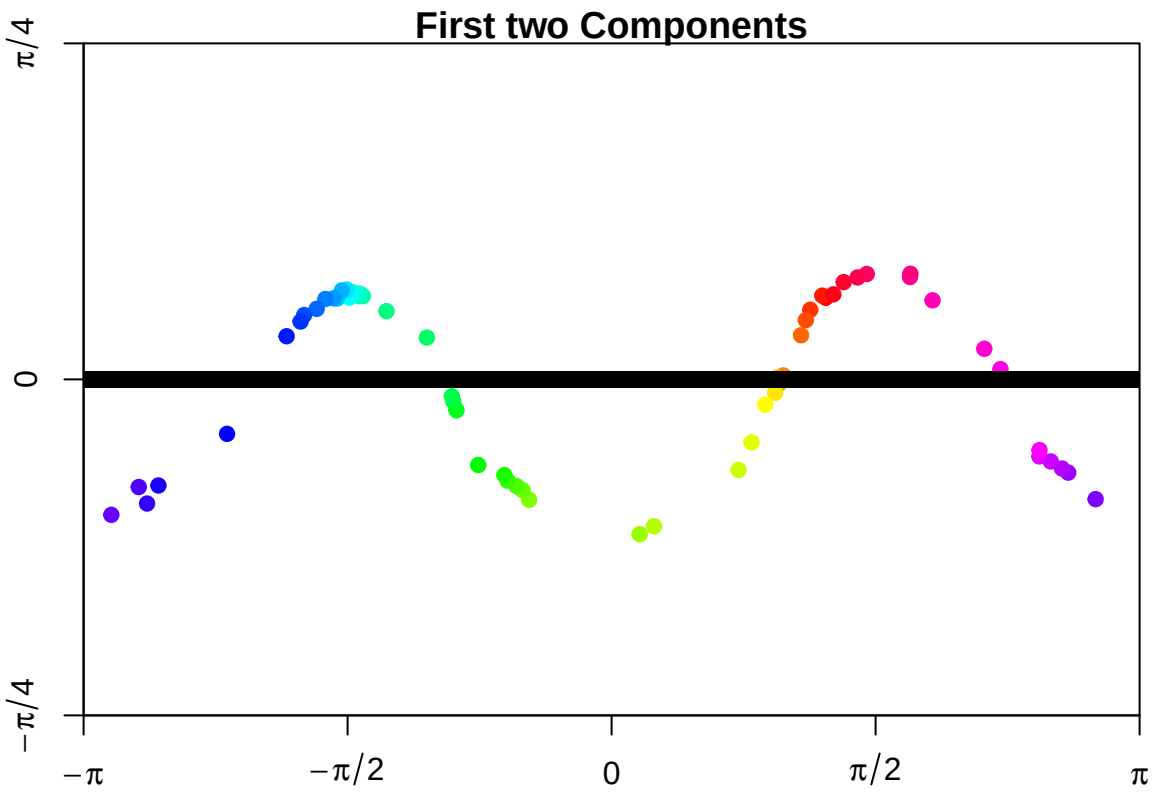

```
clusters <- cbind(samp_3[, , 1],samp_3[, , 2])

for (i in c(2:nspheres)) {
  clusters <- cbind(clusters,rotasym::r_vMF(n,c(rotasym::r_unif_sphere(1,3)),k))

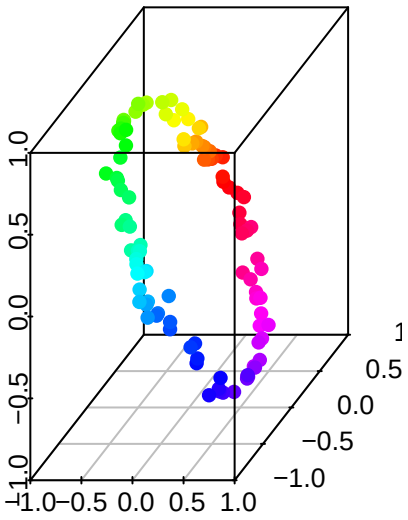
}
plot_MDS(clusters, rep(2,nspheres), angular = TRUE)
```

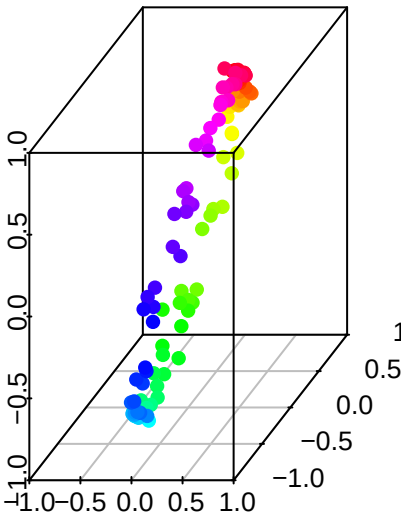

Geodesic mean
Max (+)ve from mean
Min (−)ve from mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5


First three Components, r* = 3.16227766016838
var 1
var 2
var 3
−0.6
−0.2
0.2
0.4
3
1
−1
−3
0.4
0.0
−0.6
−3
−2
−1
0
1
2
3
−0.4
0.0
0.2
0.4




```
#' # Small circle trends on (S^2)^20, almost all variance along first two spheres.
set.seed(20)
n <- 100
nspheres <- 20
k<- 500
samp_3 <- r_path_s2r(n=n, r = 2, sigma = 0.05)
layout(matrix(c(1, 2), 1))
scatterplot3d::scatterplot3d(
  samp_3[, , 1],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
scatterplot3d::scatterplot3d(
  samp_3[, , 2],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
```

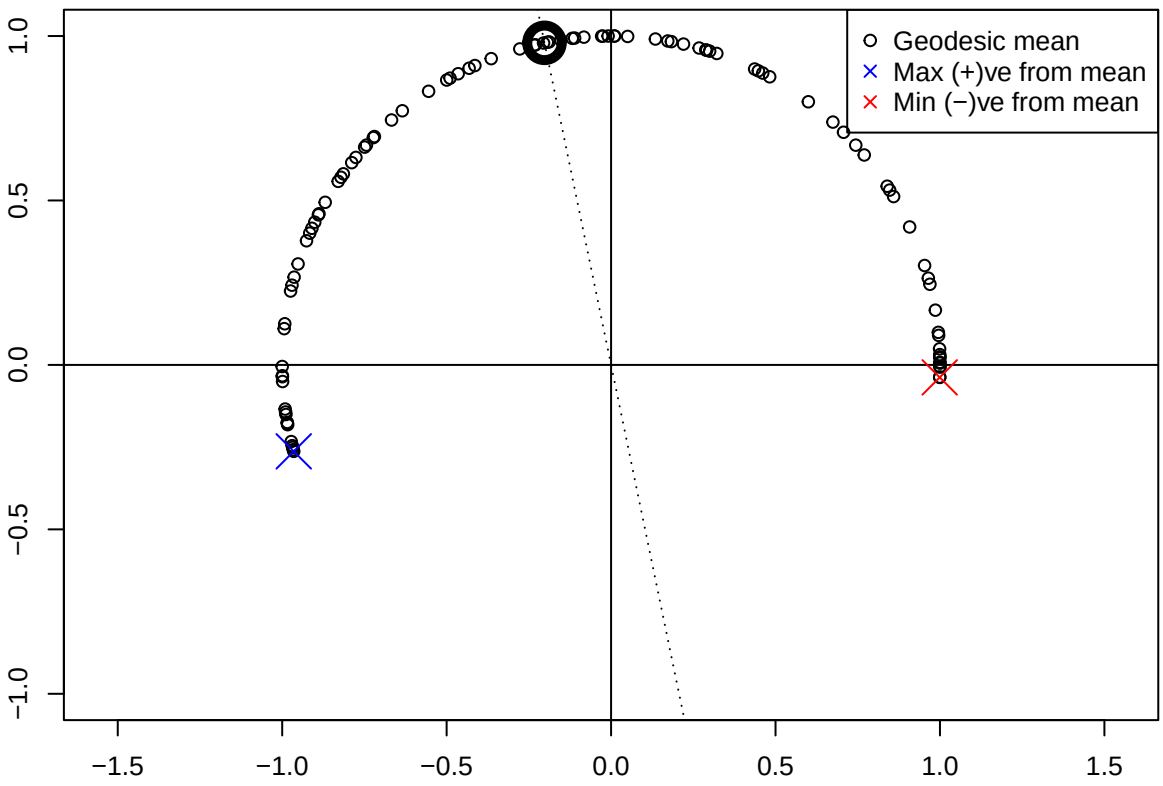


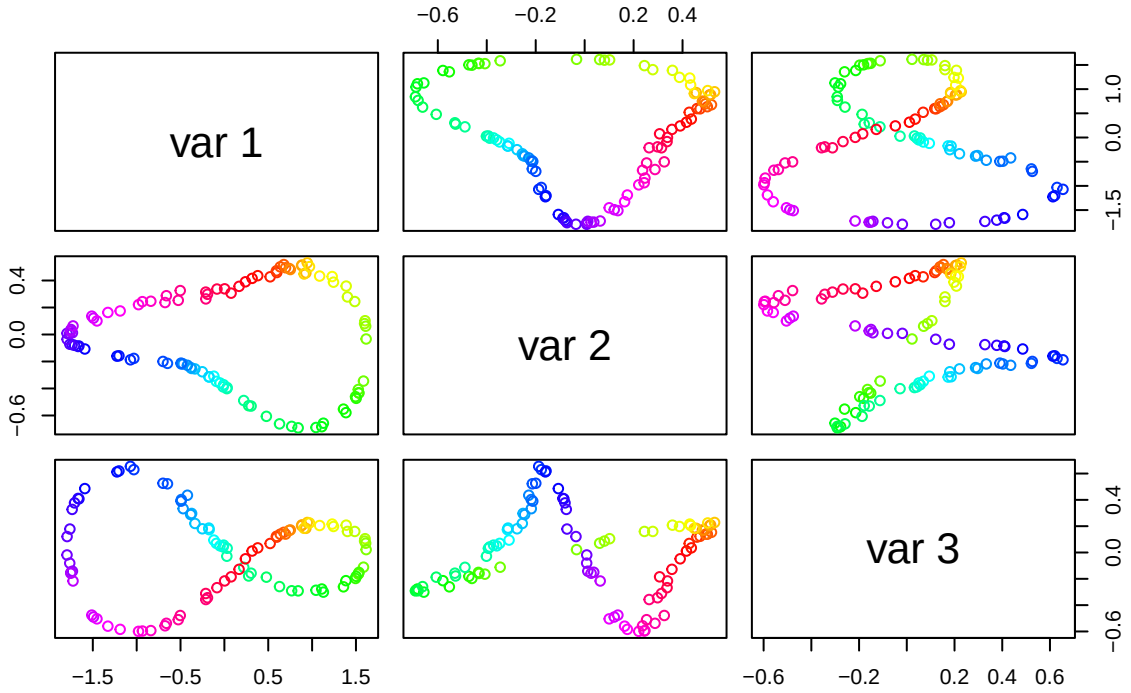


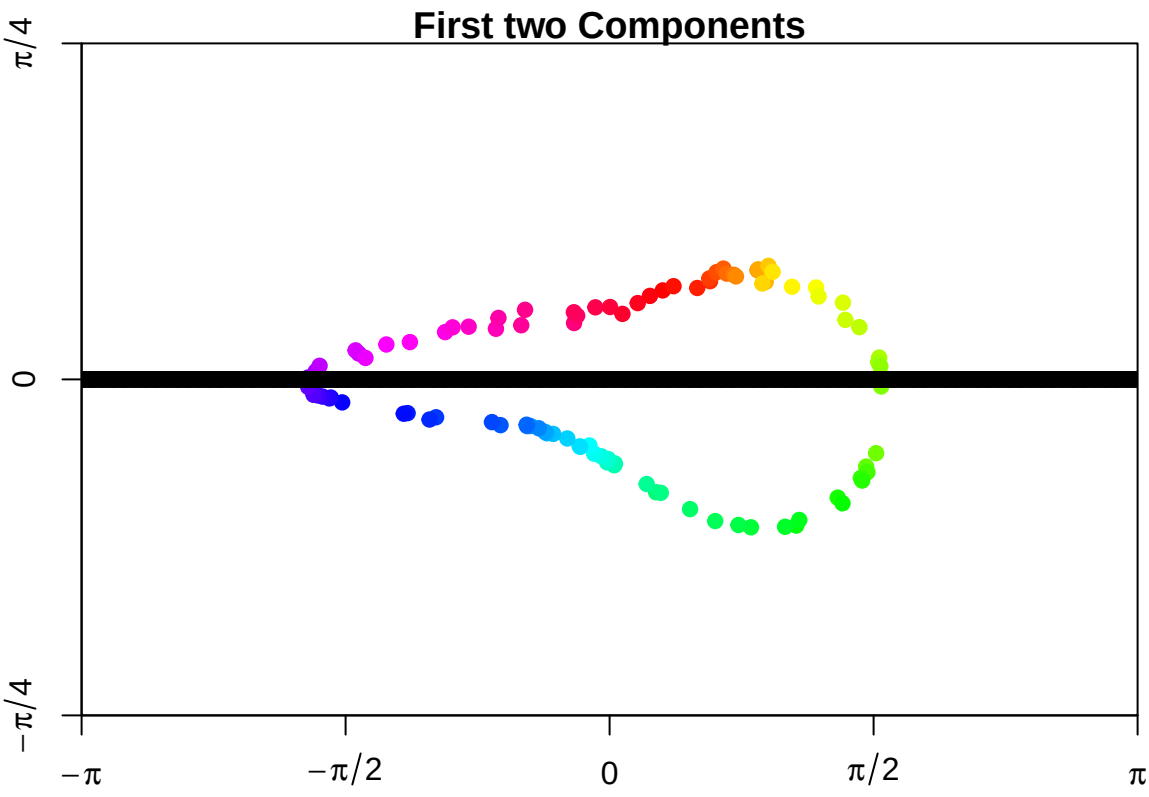

```
clusters <- cbind(samp_3[, , 1],samp_3[, , 2])

for (i in c(2:nspheres)) {
  clusters <- cbind(clusters,rotasym::r_vMF(n,c(rotasym::r_unif_sphere(1,3)),k))

}
plot_MDS(clusters, rep(2,nspheres), angular = TRUE)
```


Geodesic mean
Max (+)ve from mean
Min (−)ve from mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5


First three Components, r* = 4.47213595499958
var 1
var 2
var 3
−0.6
−0.2
0.2
0.4
−1.5
0.0
1.0
0.4
0.0
−0.6
−1.5
−0.5
0.5
1.5
−0.6
−0.2
0.2
0.4
0.6


First two Components
π/4
0
−π/4
−π
−π/2
0
π/2
π